

Protocol for the Quality-Focused Systematic Mapping Study

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Chapter 1

Protocol for the Quality-Focused Systematic Mapping Study

1.1 Protocol Identification and Status

This appendix specifies the protocol for a focused systematic mapping study with structured extraction and synthesis of quality evidence in LLM-based UML diagram generation. The study maps primary research on the generation, transformation, completion, repair, refinement, or revision of UML diagram content from textual software requirements and explicit domain descriptions, and analyzes extractable evidence about syntactic quality, semantic quality, syntax-semantic dissonance, evaluation references, measurement procedures, pragmatic adequacy, and correction effort.

The protocol follows the planning, search, selection, extraction, and synthesis principles established for secondary studies in Software Engineering Kitchenham and Charters [2007], Brereton et al. [2007], Petersen et al. [2015], Zhang et al. [2011]. Reporting will be checked against the Software Engineering Guidelines for Reporting Secondary Studies, PRISMA 2020, and PRISMA-S Kitchenham et al. [2023], Page et al. [2021], Rethlefsen et al. [2021]. Protocol amendments will be versioned and documented according to the transparency principles of PRISMA-P Shamseer et al. [2015].

Table 1.1: Protocol identification

Field	Specification
Review title	Large Language Models for UML Diagram Generation: A Focused Systematic Mapping Study of Quality Evidence
Review type	Focused systematic mapping study with structured extraction and synthesis of quality evidence
Protocol version	1.8, screening gate redesign draft (amendment A004)
Protocol date	2026-08-11
Review team	[Names, affiliations, and ORCID identifiers]
Methodological supervisor	[Name and affiliation]
Domain adjudicator	[Name and affiliation]
Registration repository	[OSF, Zenodo, or institutional repository identifier]
Planned search interval	January 2022 to the final search date
Language of eligible reports	English
Primary disciplinary domains	Software Engineering, Requirements Engineering, Software Modeling, Model Driven Engineering, and Large Language Models
Protocol status	Prospective protocol. Selection, extraction, and synthesis must not begin before protocol validation and registration, except for explicitly designated pilot activities.

1.2 Rationale and Need for the Secondary Study

Recent primary studies indicate that LLMs can generate UML diagrams that satisfy at least some criteria of textual validity or syntactic conformity while presenting problems of semantic correctness or completeness relative to source requirements and domain information De Bari et al. [2024], Ferrari et al. [2024], Garaccione et al. [2025a,b], Xiao et al. [2025]. These studies use different diagram types, source specifications, model families, prompting strategies, evaluation references, units of analysis, metrics, and terminology. Consequently, apparently similar labels such as syntactic correctness, semantic correctness, completeness, consistency, hallucination, or understandability may denote non equivalent constructs and procedures.

The conceptual background on model quality distinguishes the relation between a model and its language, the relation between a model and the relevant domain, and the relation between a model and its interpreters or uses Lindland et al. [1994], Krogstie et al. [2006]. The UML specification also distinguishes concrete representations from the abstract syntax and well formedness constraints of the modeling language Object Management Group [2017]. These distinctions are necessary for evaluating generated diagrams, but the manner in which recent LLM studies operationalize them remains to be systematically reconstructed.

Related secondary studies provide important but different forms of coverage. Reviews of UML model quality predate the current LLM based generation setting Genero et al. [2011]. Reviews of transformations from natural language to UML mainly cover rule based and earlier machine learning approaches Ahmed et al. [2022]. Recent studies map LLM applications across Model Driven Engineering or review UML and entity relationship diagram generation at a broad level Zhang et al. [2027], Rahmanian et al. [2026]. The present study has a narrower and more analytically intensive objective. It maps LLM-based UML generation studies that report extractable quality evidence and synthesizes how they define, classify, measure, and validate diagram quality and inadequacies.

The review is necessary because the subsequent thesis stages depend on construct clarity. Before the incidence of inadequacies can be estimated, their categories and units must be defined. Before contextual determinants can be investigated, the dependent quality constructs must be operationalized. Before pragmatic effects or rework can be evaluated, the literature must be examined for existing tasks, measures, and procedures. The secondary study therefore has both descriptive and constitutive functions. It maps the field and establishes the evidence base required to formulate later empirical questions and instruments.

1.3 Study Design and Analytical Focus

Systematic maps are primarily designed to structure a research area through classification and counting, whereas systematic reviews place greater emphasis on focused evidence synthesis Petersen et al. [2015]. This study combines descriptive mapping with structured extraction and synthesis of quality evidence in a single eligible corpus. Every included report must provide extractable evidence about the quality of generated UML content. Analytical groupings are derived from extraction fields, including the availability of syntactic evidence, semantic evidence, syntax-semantic dissonance evidence, and PlantUML generation; they are not separate eligibility strata.

1.4 Objective and Expected Contribution

The general objective is to map primary studies that use LLMs to generate, transform, complete, repair, refine, or revise UML diagram content from textual software requirements or explicit textual domain descriptions, to synthesize how these studies characterize, classify, operationalize, measure, validate, and report the syntactic, semantic, and pragmatic quality of the generated

UML content, the inadequacies they report, and the evidence of syntax-semantic dissonance, and to establish the quality constructs, deviation categories, units of assessment, evaluation references, reliability practices, baselines, and correction effort measures required to formulate the empirical questions and instruments of the subsequent thesis stages.

The study is expected to produce six integrated contributions.

- i) a structured map of publication, study, diagram, task, input, model, context, output, and evaluation characteristics;
- ii) a traceable matrix connecting quality terms to definitions, theoretical references, evaluation references, units, indicators, metrics, and procedures;
- iii) a normalized taxonomy of inadequacies reported in the literature, while preserving the original terminology used by each primary study;
- iv) a catalogue of semantic references, model oracles, human evaluation procedures, automated checks, and reliability practices;
- v) a synthesis of reported contextual conditions, pragmatic effects, and correction or rework measures;
- vi) a specification of the quality constructs, deviation categories, units of assessment, evaluation references, baselines, and correction effort measures for which the literature provides precedent, together with an explicit record of those for which it does not, as the basis for the empirical instruments of the subsequent thesis stages.

The review does not claim that the resulting taxonomy exhausts every possible inadequacy of every UML diagram type. It synthesizes inadequacies that are reported, operationalized, or inferable from eligible primary studies under an explicit coding procedure.

1.5 Research Questions

The overarching question is the following.

ORQ. How have primary studies used LLMs to generate, transform, complete, repair, refine, or revise UML diagram content from textual software requirements or explicit textual domain descriptions, under which generation context, and how do they characterize, classify, operationalize, measure, validate, and report the syntactic, semantic, and pragmatic quality of the generated UML content, its reported inadequacies, and the evidence of syntax-semantic dissonance?

The question is decomposed into mapping questions and synthesis questions.

1.5.1 Mapping questions

- MQ1.** How is the research landscape distributed by publication year, venue, publication type, research method, contribution type, author affiliation, and availability of replication material?
- MQ2.** Which UML diagram types, generation, transformation, completion, repair, refinement, or revision tasks, textual input forms, application domains, datasets, and output representations, including PlantUML, are investigated?
- MQ3.** Which LLM families, model versions, access modes, prompting and context strategies, demonstrations, retrieval mechanisms, fine tuning procedures, agent structures, tool integrations, feedback mechanisms, and inference parameters are reported?

- MQ4.** Which evaluation dimensions, baselines, human roles, automated procedures, and open science practices are reported across the mapped studies?
- MQ5.** How do diagram types, generation configurations, and evaluation practices co occur across the literature, and which combinations remain under investigated?

1.5.2 Quality evidence and synthesis questions

- SQ1.** Which quality constructs are used, how are they named and defined by the primary studies themselves, and how can the reported terms be normalized across the syntactic, semantic, and pragmatic dimensions without erasing study specific distinctions?
- SQ2.** Which inadequacies are reported, how are they defined, which UML elements or relations carry them, and how can the original categories be normalized without erasing diagram specific distinctions?
- SQ3.** Which metrics, rubrics, oracles, semantic references, units of assessment, scales, thresholds, aggregation rules, evaluators, automated tools, and judgment procedures are used?
- SQ4.** To what extent does the reported evidence distinguish syntactic quality from semantic quality, and what direct or indirect evidence of syntax-semantic dissonance is available?
- SQ5.** How do the studies establish the reliability and validity of quality assessments, including evaluator training, independent coding, agreement measurement, adjudication, repeated generation, sensitivity analysis, and reporting of uncertainty?
- SQ6.** How is the knowledge required for generation supplied and structured as generation context, with respect to the domain of the specified system and to the technical contract of the UML notation?
- SQ7.** How are pragmatic adequacy, inspection, correction, rework, or productivity effects defined and measured, and with what inferential strength are they claimed?

1.5.3 Traceability between questions, data, and synthesis

Table 1.2: Traceability from research questions to extracted data and synthesis products

RQ	Required data		Synthesis dure	proce-	Primary product
MQ1	Bibliographic data, study contribution, affiliation, availability	meta-design, type, artifact	Frequencies, trends, venue classification, and cross tabulation	temporal classification	Publication and evidence landscape
MQ2	Diagram type, task, input, domain, dataset, output representation		Classification and occurrence analysis	co	Diagram, task, and data map

RQ	Required data	Synthesis procedure	proce-	Primary product
MQ3	Model, version, access, context, prompt, retrieval, adaptation, tools, parameters	Frequency distributions and configuration profiles	distribu-	Technical configuration map
MQ4	Evaluation dimensions, baselines, evaluators, automation, artifacts	Descriptive classification and cross tabulation		Evaluation and reproducibility map
MQ5	Combined categories from MQ2 to MQ4	Co occurrence matrices and gap analysis		Research concentration and gap map
SQ1	Original terms, definitions, citations, operational definitions, examples	Conceptual comparison and normalized construct matrix		Quality construct matrix
SQ2	Original inadequacy labels, textual definitions, examples, affected UML carrier, reference violated	Hybrid deductive and inductive coding, constant comparison, and negative case analysis		Taxonomy of reported inadequacies
SQ3	Reference, oracle, unit, formula, denominator, scale, threshold, aggregation, procedure	Metrological comparison and traceability analysis		Evaluation reference and metric catalogue
SQ4	Syntactic and semantic evidence availability, dissonance classification, evidence basis	Extraction-based comparison and evidence grouping		Syntax-semantic dissonance evidence map
SQ5	Reviewer roles, training, independence, agreement, adjudication, repetitions, uncertainty	Methodological appraisal and reliability profile		Assessment credibility matrix
SQ6	Domain knowledge supplied and UML technical instruction supplied	Classification and cross tabulation over the whole corpus		Generation context knowledge map
SQ7	Context factor, inferential status, task, actor, outcome, correction and rework measures	Narrative synthesis stratified by inferential strength		Pragmatic and rework evidence map

1.6 Conceptual and Operational Scope

1.6.1 Population and units

The documentary population comprises scientific reports published within the defined time interval that investigate LLM based generation of UML diagrams from textual software requirements or explicit textual domain descriptions. Three units are distinguished.

- a) the **report**, which is the document screened for eligibility;

- b) the **primary study**, which is the empirical or technical investigation that may be reported in one or more publications;
- c) the **evaluation instance**, which is an extractable combination of diagram type, input corpus, model configuration, generation condition, and evaluation procedure.

The report is the unit of selection. The primary study is the unit used to prevent double counting. The evaluation instance is the preferred unit for analytical extraction when a report contains multiple diagram types, models, treatments, datasets, or evaluation procedures.

1.6.2 Operational definitions

Table 1.3: Operational definitions used in the protocol

Term		Operational definition
Large Language Model		A neural language model identified by the study as a large, foundation, generative, instruction following, or conversational language model and used as a substantial component of the generation process. Rule based NLP and conventional machine learning without an LLM are outside scope. The operational boundary in Section 1.9.4 governs cases in which a report uses broad labels such as language model, pretrained language model, or foundation model without clearly establishing a generative or semantically substantive role in UML content production.
UML diagram		A model or diagram explicitly identified as UML or as one of the UML diagram types. PlantUML output is eligible only when it is intended to encode UML. Generic Mermaid, C4, ER, BPMN, or architectural diagrams are outside scope unless the UML component is separable.
Textual software requirements		Natural language or structured textual descriptions of intended software behavior, structure, goals, actors, rules, constraints, scenarios, or domain information. Examples include requirements documents, use case narratives, user stories, problem statements, feature descriptions, and domain descriptions.
Generation task		Production, completion, transformation, repair, refinement, or iterative revision of a UML diagram in which the LLM materially determines the resulting model content.
Generation context		The complete information made available to the model for an evaluation instance, including instructions, requirements, domain information, UML knowledge, examples, retrieved material, tools, prior turns, and ordering.
Textual validity		Conformance of the concrete output string to the grammar accepted by the selected representation or rendering tool. It does not by itself establish UML conformity.
UML syntactic conformity		Conformance of the represented model to the applicable UML abstract syntax, admissible constructs, relations, and well formedness constraints.
Semantic correctness		Degree to which represented propositions, elements, relations, constraints, states, events, or behaviors are supported by the relevant requirements and domain reference.

Term	Operational definition
Semantic completeness	Degree to which relevant units in the defined semantic reference are adequately represented in the diagram.
Pragmatic adequacy	Degree to which the diagram supports a defined engineering activity for a specified actor and context. Comprehensibility may be one component but is not treated as a sufficient synonym.
Inadequacy	A reported or inferable discrepancy between the generated diagram and an explicit language, semantic, or task reference. The review preserves the source term and separately assigns a normalized code.
Semantic reference	The source against which correctness or completeness is assessed, such as requirements, explicit domain knowledge, an annotated reference model, a validated model, a gold standard, a rubric, or expert judgment.
Correction or rework	Human or automated activity required after initial generation to inspect, change, validate, or regenerate a diagram so that it satisfies a defined acceptance criterion.
Publication family	Two or more reports that describe the same or substantially overlapping primary study, dataset, experiment, or artifact.

1.6.3 Temporal, language, and publication scope

The planned publication interval begins in January 2022 and ends on the final search date. The start year reflects the emergence of current instruction following and conversational LLM use in Software Engineering and will be checked during search validation. If the seed set identifies an eligible study before 2022, the interval will be extended and the amendment documented.

Reports written in English are eligible. Journal articles, conference papers, and workshop papers are eligible when they provide a complete scientific report. Preprints are eligible because of the rapid evolution of the research area, but publication status is coded explicitly and peer reviewed and preprint evidence are analyzed separately when the distinction affects interpretation. Abstracts, posters, slide decks, editorials, prefaces, tutorials, theses, books, patents, and non scientific web content are excluded from the primary corpus. Secondary and tertiary studies are excluded from the primary corpus but retained for background, search validation, and snowballing.

1.7 Information Sources

The identification strategy combines broad bibliographic indexes, direct publisher or professional society libraries, a preprint repository, citation searching, and manual venue inspection. These source types perform complementary functions and are not interchangeable. Broad indexes reduce dependence on a predefined venue list, direct libraries improve access to conference and workshop proceedings, arXiv captures rapidly disseminated preprints, and manual venue inspection verifies coverage gaps rather than defining the documentary population.

The source set is informed by methodological guidance on complementary search routes in Software Engineering secondary studies Brereton et al. [2007], Zhang et al. [2011], Wohlin [2014] and by recent secondary studies showing that LLM based modeling research is distributed across main conferences, companion and workshop tracks, and journals Zhang et al. [2027], Rahmanian et al. [2026]. These prior studies provide empirical input to the initial source configuration, but the final configuration remains subject to seed set retrieval and source contribution analysis during protocol validation.

Table 1.4: Planned information sources, decision status, and functions

Source	Status	Function	Search treatment
Scopus	Core	Broad multidisciplinary index	Title, abstract, and keyword search with complete export and source provenance preservation
Web of Science Core Collection	Core	Independent multidisciplinary index and citation data	Topic search with complete export; also used for forward citation discovery when available
IEEE Xplore	Core	Direct coverage of IEEE and IEEE/ACM venues	Metadata search adapted to platform syntax, with complete export of the available bibliographic fields
ACM Digital Library	Core	Direct coverage of ACM venues	ACM Full-Text Collection; validated source-specific query restricts the LLM block to title, abstract, or author keywords while retaining UML and generation/input in the platform’s broad Anywhere field; complete candidate export and proceedings/workshop coverage inspection
ScienceDirect	Provisional complementary	Direct Elsevier publisher coverage	Retained when the validation pilot identifies unique eligible or seed records, or a documented coverage gap not resolved by Scopus or Web of Science; otherwise its exclusion is justified before protocol registration
arXiv	Complementary	Recent preprints and early dissemination	Search with the same conceptual blocks; preprint and subsequently published versions are linked as one publication family
Google Scholar	Citation discovery only	Broad forward citation discovery	Used during snowballing, not as a principal automated database because ranking, result limits, and export behavior reduce reproducibility
Manual venue search	Coverage validation	Recent indexing gaps and direct inspection of recurrent specialist outlets	Proceedings and journal tables of contents are inspected only for venues validated by the criteria below; this route does not replace the automated database search

1.7.1 Role of automated sources and manual venue inspection

The automated database search is the primary mechanism for defining the candidate documentary corpus. The manual venue search is a complementary coverage validation mechanism. Consequently, the conclusions of the review are not restricted to a predetermined set of conferences or journals, and a study remains eligible regardless of publication venue when it satisfies the review criteria.

The initial core for manual inspection is shown in Table 1.5. The list reflects the intersection of Software Modeling and Model Driven Engineering, Requirements Engineering, LLM based Software Engineering, and empirical evaluation. It also incorporates outlets recurrently identified in recent secondary studies and venues that contain known seed studies on UML diagram generation by LLMs Zhang et al. [2027], Rahmanian et al. [2026], De Bari et al. [2024], Ferrari et al. [2024], Garaccione et al. [2025a,b], Xiao et al. [2025].

A venue is promoted from the conditional set to manual inspection when at least one of the following conditions holds.

- a) the validated seed set contains a relevant study from the venue that is not retrieved by any core automated source;
- b) at least two eligible candidate studies are identified in the venue during automated searching, pilot screening, or snowballing;
- c) methodological and domain reviewers jointly determine that excluding the venue creates a documented coverage risk.

The venue decision log records the evidence for inclusion, the years inspected, the search

Table 1.5: Initial manual venue inspection set

Status	Venues	Rationale and treatment
Core	MODELS main conference; MODELS Companion; MODELSWARD; <i>Software and Systems Modeling</i>	Central outlets for software modeling, UML, and Model Driven Engineering. Their proceedings or tables of contents are inspected for every year in the review interval.
Core	IEEE International Requirements Engineering Conference and directly related workshops; REFSQ	Relevant to transformation from textual requirements and domain descriptions into software models.
Core	ICSE main conference; ICSE Software Engineering in Practice; directly relevant ICSE tracks or workshops; NLBSE	Relevant to LLM based Software Engineering, industrial evaluation, natural language, and generation tasks.
Core	ESEM	Relevant to empirical evaluation, measurement, comparison, and quality assessment.
Conditional	EASE and <i>Empirical Software Engineering</i>	Added to manual inspection when the seed set or automated results demonstrate recurrent eligible studies or an indexing gap.
Conditional	ECMFA and associated STAF events; SLE	Added when model transformation, language engineering, or formal modeling studies are recurrent in the candidate corpus.
Conditional	ASE; FSE; <i>Automated Software Engineering</i> ; <i>Journal of Systems and Software</i> ; <i>Requirements Engineering</i> ; IEEE TSE; ACM TOSEM	Added when at least two eligible studies are identified in the outlet, or when a known eligible seed study is not retrieved by the automated sources.

route, and the number of unique eligible records contributed. Manual inspection is not expanded solely because a venue is prestigious or generally relevant to Software Engineering.

1.7.2 Manual venue inspection procedure

Manual venue inspection is conducted as a census-like inspection of predefined documentary units rather than as a repetition of the automated Boolean query inside each venue. Its purpose is to identify potentially relevant reports that may be missed because of terminology variation, indexing delay, incomplete database coverage, or publication in companion and satellite tracks. The manual route therefore prioritizes sensitivity during discovery and applies the formal eligibility criteria only after records have been merged into the master corpus.

1.7.2.1 Operational unit and source hierarchy

The operational unit of manual inspection is defined as

$$U_m = \langle \text{venue}, \text{year}, \text{volume}/\text{track} \rangle.$$

For conference families, main proceedings and companion proceedings are treated as distinct inspection units when published separately. Workshops, doctoral symposia, tools and demonstration tracks, educator tracks, industry tracks, and other satellite events are not excluded a priori when their reports form part of the inspected proceedings. Journal inspection is performed by volume and issue within the review interval.

For every unit, the documentary source is selected according to the following order of preference.

- 1) official publisher proceedings or journal table of contents;
- 2) official conference or journal table of contents that links to the definitive records;
- 3) official program pages used only as navigation support when the publisher table of contents is incomplete or difficult to traverse.

Aggregator pages, unofficial bibliographies, and general web search results are not used as the sole evidence that a manual-search unit has been completely inspected.

1.7.2.2 Source roles, documentary membership, and provenance chain

The manual-search evidence model distinguishes documentary membership from metadata enrichment. For each unit U_m , every source is assigned one of four operational roles.

- **PRIMARY_TOC**: the authoritative publisher proceedings, journal table of contents, or equivalent official publication inventory that establishes the initial documentary membership of the unit;
- **VENUE_CROSSCHECK**: an independent official venue or journal source used to verify completeness and identify discrepancies in the primary inventory;
- **METADATA_SOURCE**: an authoritative bibliographic source, publisher record, Crossref, or equivalent service used to enrich records already identified in the documentary universe;
- **AUXILIARY_NAVIGATION**: an official navigation aid that facilitates traversal but does not independently establish documentary membership.

Only the **PRIMARY_TOC** establishes initial membership. A cross-check or metadata source may reveal an omitted, duplicated, withdrawn, reclassified, or otherwise discrepant record, but that record is not silently incorporated into the final documentary universe. The discrepancy is first recorded as an **InventoryConflict** and explicitly reconciled against the authoritative publication evidence. This rule prevents metadata services or search engines from redefining the population being manually inspected.

The provenance chain for each manual-search unit is operationalized as follows.

```
PRIMARY_TOC / permissible source snapshot
→ source/source_manifest.csv
→ raw/inventory_raw.csv
→ normalized/inventory.csv
→ screening/discovery.csv
```

This chain is retained in the replication package so that an external auditor can determine which source established the unit, which documentary items were observed, how bibliographic values were normalized or enriched, and whether every documentary item was subsequently inspected. This implementation follows the transparency and reproducibility objectives of systematic-search reporting Rethlefsen et al. [2021] and the traceability requirements expected of systematic mapping studies in Software Engineering Petersen et al. [2015].

Source manifest. Each unit maintains a `source_manifest.csv` containing, at minimum, `SourceID`, `ManualSearchUnitID`, `SourceRole`, `URL`, `RetrievedAt`, `LocalSnapshot`, `SHA256`, `ContentType`, and `Notes`. A local snapshot is preserved only when access conditions and redistribution rights permit it. When a snapshot cannot be redistributed, the replication package preserves the source URL, retrieval timestamp, permitted audit metadata, and a locally computed checksum when applicable without redistributing restricted source content.

Raw documentary inventory. The file `raw/inventory_raw.csv` materializes every item observed in the documentary universe established by the **PRIMARY_TOC**, irrespective of relevance. Its minimum schema is `ManualSearchID`, `ManualSearchUnitID`, `SourceOrdinal`, `InventorySourceID`, `SourceRecordLocator`, `TitleRaw`, `AuthorsRaw`, `DOIRaw`, `VenueRaw`, `YearRaw`, `VolumeTrackIssueRaw`, `PublisherRecordURLRaw`, `RetrievedAt`, `ExtractionMethod`, and `Notes`. Values are retained as observed in the source. `ManualSearchID` is assigned when the documentary item is materialized in this raw inventory and remains stable across all subsequent layers.

Normalized inventory. The file `normalized/inventory.csv` preserves the raw values while adding normalized or enriched representations such as normalized title, authors, DOI, publisher metadata, abstract, author keywords, and full-text location. Every normalized record must be traceable to an existing `ManualSearchID` in the raw inventory. Metadata enrichment may repair formatting, resolve an identifier, or add bibliographic fields, but it does not silently create documentary membership. A temporary reconciliation record may be retained while a discrepancy is investigated, but it remains explicitly conflict-marked and cannot contribute to the final unit membership until the conflict has been resolved.

Immutability and normalization. Once the raw inventory for a documented extraction event has been committed to the replication package, observed values are not silently overwritten. Corrections, canonicalization, and metadata augmentation occur in the normalized layer, preserving the raw evidence that motivated the transformation. When the authoritative source itself changes after retrieval, the later state is captured as a new source retrieval event rather than replacing the previously recorded evidence.

1.7.2.3 Inventory and inspection sequence

Before any relevance decision is made, the reviewers materialize the complete documentary inventory in `raw/inventory_raw.csv`. The raw inventory contains every item belonging to the universe established by the `PRIMARY_TOC`, including records that will later be classified as clearly irrelevant or as non-research items. No candidate filtering is permitted during this stage. The inventory preserves the observed order and source locator whenever available, and assigns the stable `ManualSearchID` used by the normalized and screening layers. Inspection then proceeds in the following order.

- 1) Inspect the title of every record in the unit.
- 2) When the title does not permit unequivocal exclusion from the discovery scope, inspect the abstract and author keywords when available.
- 3) Mark as a manual-search candidate any report for which there is a reasonable possibility that an LLM plays a substantive role in generating, transforming, completing, repairing, or iteratively revising UML or an explicitly identified UML diagram type from textual software requirements or explicit textual domain information.
- 4) Defer ambiguous Layer 1 eligibility judgments to the formal screening stage rather than resolving them during manual discovery.
- 5) Preserve the provenance of every candidate even when the same report is later found in an automated source.

The automated search string is not reapplied as a mandatory manual-search filter. Terms such as LLM, ChatGPT, UML, class diagram, sequence diagram, requirements, model generation, and related expressions are used as attention cues during inspection, but the absence of those literal terms from the title does not justify exclusion when the abstract suggests potential relevance. This decision is intended to reduce dependence on terminology already encoded in the automated query.

1.7.2.4 Manual-search discovery statuses

Manual discovery uses the provisional statuses in Table 1.6. These statuses are discovery annotations and do not replace the Layer 1 inclusion and exclusion criteria.

Table 1.6: Manual-search discovery statuses

Status	Operational meaning
CLEAR-CANDIDATE	LLM use, UML or an explicitly identified UML diagram type, and a generation, transformation, completion, repair, or revision task are sufficiently explicit to justify candidate registration.
POSSIBLE-CANDIDATE	Relevance is plausible, but one or more elements such as the UML status of the notation, textual input, or substantive role of the LLM require full-text screening. The report is retained as a candidate.
CLEAR-NON-CANDIDATE	Title and, when required, abstract provide sufficient evidence that the report is incompatible with the discovery scope. The record contributes to the inspected-item count but is not promoted to the candidate set.
NON-RESEARCH-ITEM	The item is a keynote, editorial, index, slide deck, poster-only item, front matter, or another publication form that does not constitute a primary scientific report. The item is counted for auditability but is not treated as a candidate primary study.

1.7.2.5 Manual-search record

After the complete raw inventory has been established, every report promoted to the manual-search candidate set receives the following discovery and screening fields before cross-source deduplication. These fields supplement rather than replace the raw and normalized inventories.

Table 1.7: Minimum manual-search record

Field	Content
ManualSearchID	Stable identifier assigned when the item is materialized in the raw documentary inventory and preserved across normalized, discovery, and screening layers.
VenueFamily	Conference or journal family, for example MODELS or Software and Systems Modeling.
VenueOrJournal	Exact publication outlet.
Year	Publication year associated with the inspected unit.
VolumeTrackIssue	Main proceedings, companion proceedings, workshop, track, journal volume, or journal issue when applicable.
SourceURL	Official publisher or official venue table-of-contents URL used for inspection.
InspectionDate	Date on which the unit was inspected.
Title	Report title as published.
Authors	Authors as published.
DOIorPID	DOI or other persistent identifier when available.
DiscoveryStatus	CLEAR-CANDIDATE or POSSIBLE-CANDIDATE for records promoted to the candidate set.
CandidateRationale	Short statement explaining why the report may satisfy the review scope.
LLMEvidence	Title, abstract, or keyword evidence indicating substantive LLM use.
ModelingEvidence	Evidence indicating UML, a UML diagram type, or potentially relevant software modeling.
InputEvidence	Evidence indicating textual requirements, user stories, specifications, scenarios, or explicit domain descriptions when visible at discovery.

Field	Content
AutomatedSourceMatch	Populated after merge to indicate Scopus, IEEE Xplore, ACM Digital Library, Web of Science, or other automated-source matches.
Layer1Decision	Populated only during the formal screening process.

1.7.2.6 Positive coverage controls

Known relevant studies that belong to an inspected venue are used as positive coverage controls for the manual-search procedure. Their role is diagnostic rather than preferential: they remain subject to the same eligibility criteria as other reports. Failure to recover a known control during complete inspection of its venue-year unit triggers reinspection of that unit and review of the inspection procedure. Examples include the known seed studies already used in automated search validation when their publication outlet belongs to the manual-search core.

1.7.2.7 Merge with automated sources and incremental yield

Manual candidates are merged with the master bibliographic corpus only after their original manual provenance has been preserved. Deduplication uses DOI, normalized title, author and year information, and publication-family reconciliation according to Section 1.10. A manual candidate that is also retrieved from an automated source remains tagged as having been found manually.

The incremental contribution of manual searching is reported using at least the following quantities.

$N_{\text{manual,cand}}$ = number of manual-search candidates,

$N_{\text{manual,eligible}}$ = number of those candidates subsequently satisfying Layer 1,

and

$N_{\text{manual,exclusive}}$ = number of Layer 1 studies not retrieved by any retained automated source.

The incremental manual-search yield is calculated as

$$Yield_{\text{manual}} = \frac{N_{\text{manual,exclusive}}}{N_{\text{manual,eligible}}}$$

when $N_{\text{manual,eligible}} > 0$. The raw number of records inspected is also reported so that manual-search effort and yield can be interpreted together.

1.7.2.8 Completion rule for a manual-search unit

A unit U_m progresses through the operational states **PENDING**, **IN_PROGRESS**, **COMPLETE**, and **BLOCKED**. **PENDING** denotes a planned unit whose inventory has not yet begun. **IN_PROGRESS** denotes an active inventory, reconciliation, or inspection process. **BLOCKED** denotes a unit whose completion is prevented by an unresolved access limitation or documentary-membership conflict.

A unit is marked **COMPLETE** only when all of the following conditions hold:

- a) the **PRIMARY_TOC** and any cross-check sources used have been recorded in the source manifest with retrieval dates and reproducibility metadata;
- b) the raw documentary inventory materializes every item in the reconciled documentary universe;

- c) every raw item has been traversed during manual discovery, including clearly irrelevant records and non-research items;
- d) every normalized record is traceable to a raw `ManualSearchID`;
- e) all material `InventoryConflict` instances have been reconciled rather than merely documented;
- f) the total item count, candidate count, discovery-status counts, inspection date, access limitations, and candidate identifiers have been recorded in the manual venue search log; and
- g) the audit artifacts required for the unit are preserved in the replication package subject to licensing and redistribution constraints.

An unresolved documentary-membership conflict prevents `COMPLETE`. A documented but unresolved conflict therefore leaves the unit `IN_PROGRESS` while reconciliation is active, or `BLOCKED` when reconciliation cannot proceed. Units whose proceedings are not yet available remain `PENDING` and are revisited during the final search update before synthesis.

Manual searching may be operationalized in temporal batches without changing the review interval or eligibility criteria. The initial operational batch prioritizes recent years and high-yield core venues, but a venue family is considered completely searched only after every required year in the protocol interval has reached `COMPLETE`. Interim analyses therefore do not redefine the final documentary scope.

1.7.2.9 Initial operational order

To maximize early coverage of the communities most directly related to the phenomenon while preserving the complete core venue set, manual inspection is initially conducted in the following order:

- 1) MODELS main conference and MODELS Companion;
- 2) IEEE International Requirements Engineering Conference and directly related workshops;
- 3) NLBSE;
- 4) ICSE main conference and Software Engineering in Practice;
- 5) ESEM;
- 6) MODELSWARD;
- 7) *Software and Systems Modeling*;
- 8) REFSQ.

This order is an operational prioritization and is not interpreted as a ranking of scientific importance. Conditional venues are activated only by the criteria specified in Table 1.5 and the associated venue decision rules.

1.7.2.10 Manual-search audit outputs

For every venue family, the replication package contains linked audit artifacts:

- 1) a *source manifest*, identifying the primary documentary source, cross-check sources, meta-data sources, retrieval times, and permissible snapshots or checksums;
- 2) a *raw documentary inventory*, materializing every item in the reconciled documentary universe before relevance classification;
- 3) a *normalized inventory*, preserving traceability to raw identifiers while recording normalized and enriched bibliographic metadata;
- 4) a *venue search and discovery log*, demonstrating which venue-year-volume units were inspected, the status assigned to every raw item, and whether each unit is complete;
- 5) a *manual candidate set*, preserving candidate records and discovery rationales before destructive deduplication; and
- 6) a *coverage comparison*, identifying which manual candidates were also found by each automated source and which eligible studies were contributed exclusively by manual inspection.

The final report summarizes the number of completed manual-search units, total records inspected, candidate reports, eligible reports, and eligible studies contributed exclusively by manual inspection. This evidence is used to evaluate the incremental value of manual venue inspection rather than merely stating that handsearching was performed.

1.7.3 Source contribution and retention analysis

Before protocol registration, the validation pilot will compare the contribution of each planned source. For every seed and pilot record, the review team records which sources retrieve it and whether the source contributes a unique eligible record. Core sources are retained because they represent distinct indexing or publisher functions. ScienceDirect remains provisional and is retained in the production search only when it retrieves at least one unique eligible or seed study, or when its removal leaves a documented publisher coverage gap. A decision to omit ScienceDirect is made before protocol registration and is reported with the validation results. Any source change after registration requires a formal protocol amendment.

1.8 Search Strategy

1.8.1 Conceptual search blocks

The search strategy is intentionally broad enough to identify candidate studies for the focused review. It is organized into three conceptual blocks.

The baseline logical form is the following.

(LLM terms)
AND (UML and diagram terms)
AND (generation and source terms)

The search does not require quality terms. Requiring terms such as semantic, correctness, or hallucination at retrieval time could exclude generation studies that evaluate quality under different terminology. Extractable quality evidence is therefore never used as an eligibility filter at any stage, and is instead classified after retrieval through structured full-text inspection on the axes of Section 1.9.5. Excluding studies that lack the review’s evaluation vocabulary, whether at retrieval or at screening, would select on the review’s own dependent variable and bias the map towards the studies that already speak the language the review exists to describe.

Table 1.8: Conceptual blocks of the search strategy

Block	Candidate terms
LLM	“large language model*”, LLM*, “generative AI”, “generative artificial intelligence”, ChatGPT, GPT, Claude, Gemini, Llama, Qwen, DeepSeek
UML and diagrams	UML, “Unified Modeling Language”, PlantUML, “class diagram*”, “use case diagram*”, “sequence diagram*”, “activity diagram*”, “state machine diagram*”, “state diagram*”, “component diagram*”, “deployment diagram*”, “object diagram*”, “communication diagram*”, “package diagram*”, “composite structure diagram*”, “timing diagram*”, “interaction overview diagram*”
Generation and source	generat*, creat*, construct*, synthes*, produc*, transform*, complet*, repair*, refine*, “natural language”, requirement*, specification*, “user stor*”, scenario*, “problem statement*”, “domain description*”

1.8.2 Generic baseline search string

```

("large language model*" OR LLM* OR "generative AI" OR
"generative artificial intelligence" OR
ChatGPT OR GPT OR Claude OR Gemini OR Llama OR Qwen OR DeepSeek)

AND

(UML OR "Unified Modeling Language" OR PlantUML OR
"class diagram*" OR "use case diagram*" OR "sequence diagram*" OR
"activity diagram*" OR "state machine diagram*" OR "state diagram*" OR
"component diagram*" OR "deployment diagram*" OR "object diagram*" OR
"communication diagram*" OR "package diagram*" OR
"composite structure diagram*" OR "timing diagram*" OR
"interaction overview diagram*")

AND

(generat* OR creat* OR construct* OR synthes* OR produc* OR transform* OR
complet* OR repair* OR refine* OR "natural language" OR requirement* OR
specification* OR "user stor*" OR scenario* OR "problem statement*" OR
"domain description*")

```

The exact string executed in each source will preserve this conceptual logic while adapting field identifiers, wildcards, phrase syntax, query length restrictions, and filters. Every executed string, date, filter, and result count will be preserved verbatim in the replication package in accordance with PRISMA-S Rethlefsen et al. [2021].

1.8.3 Search validation

Search validation is performed before the production search in each source Zhang et al. [2011]. Validation is treated as an auditable sequence of controlled modifications rather than as an informal refinement of the query. Whenever possible, a single lexical or field decision is changed per round so that differences in retrieval can be attributed to the modified component.

The general validation procedure comprises the following steps.

- 1) Construct a prespecified seed set of known relevant primary studies covering different UML diagram types and publication venues.
- 2) Execute the baseline search and verify the recovery of every seed study.
- 3) Diagnose retrieval noise and potential false negatives by inspecting differential records, including title, abstract, author keywords, database indexing terms, and publication meta-data.
- 4) Modify terms or fields only when the modification has a conceptual justification and its effect can be evaluated against the preceding round.

- 5) Preserve the complete result export from every validation round.
- 6) Record the database-reported result count, the number of unique record identities used for diagnostic comparison, seed recovery, the records added or removed, the interpretation of those differences, and the resulting decision.
- 7) Re-execute the validation until the source-specific strategy preserves the prespecified seeds and no tested modification provides a better recall–precision trade-off for the scope of the study.
- 8) Compare source contribution after the source-specific strategies have been validated, recording seed retrieval, unique eligible records, and documented coverage gaps.
- 9) Validate the core and conditional manual venue sets using the predefined promotion criteria and record the years inspected.
- 10) Submit the final multi-source strategy to one reviewer with expertise in secondary studies and one reviewer with expertise in UML or LLM-based modeling.

1.8.3.1 Prespecified Scopus seed set

The first source-specific validation was conducted in Scopus on 10 August 2026. Five primary studies were prespecified as the core seed set. The seed set is a validation instrument and not a privileged inclusion list; every seed study is subject to the same eligibility criteria as every other retrieved report.

Table 1.9: Prespecified seed studies used in the Scopus validation

ID	Study	DOI
SE1	De Bari et al., <i>Evaluating Large Language Models in Exercises of UML Class Diagram Modeling</i> De Bari et al. [2024]	https://doi.org/10.1145/3674805.3690741
SE2	Ferrari et al., <i>Model Generation with LLMs: From Requirements to UML Sequence Diagrams</i> Ferrari et al. [2024]	https://doi.org/10.1109/REW61692.2024.00044
SE3	Garaccione et al., <i>A Comparison of Different Large Language Models for the Generation of UML Class Diagrams</i> Garaccione et al. [2025a]	https://doi.org/10.1109/MODELS-C68889.2025.00078
SE4	Garaccione et al., <i>Evaluating Large Language Models in Exercises of UML Use Case Diagrams Modeling</i> Garaccione et al. [2025b]	https://doi.org/10.1109/NLBSE66842.2025.00015
SE5	Xiao et al., <i>UML Sequence Diagram Generation: A Multi-Model, Multi-Domain Evaluation</i> Xiao et al. [2025]	https://doi.org/10.1109/ICSE-SEIP66354.2025.00030

All five prespecified seed studies were retrieved in every Scopus validation round. Consequently, seed recovery was $5/5 = 100\%$ for S1, S2, S3, and S4. Because the five seeds use comparatively explicit UML terminology, seed recovery was not treated as sufficient evidence by itself. Differential-record inspection was additionally used to evaluate whether a query modification removed substantively relevant studies whose UML relationship was expressed only in author keywords or less explicit metadata.

1.8.3.2 Scopus validation rounds

The initial Scopus query searched all three conceptual blocks in **TITLE-ABS-KEY**. Four controlled rounds were executed. The raw counts in Table 1.10 are the counts reported by Scopus and preserved in the exported RIS files. For diagnostic set comparison, records were also matched by DOI when available and otherwise by normalized title. Four duplicate record identities were present in each export; these duplicates do not affect the between-round differences and remain subject to the production deduplication procedure specified in Section 1.10.

Table 1.10: Scopus search-strategy validation log

Round Controlled modification		Scopus N	Seeds	Interpretation and deci- sion
S1	Baseline. All three conceptual blocks searched in TITLE-ABS-KEY ; the LLM block included “foundation model*”.	376	5/5	High seed recovery, but differential inspection identified precision loss from the polysemy of “foundation model*” and from UML matches attributable only to Scopus indexing terms. The baseline was not frozen.
S2	Removed only “foundation model*”; all fields otherwise identical to S1.	371	5/5	Five records were removed. Inspection of all five showed that none was eligible for the present scope, including pre-LLM literature in which “foundation model” had a different meaning. The lexical modification was accepted.
S3	Restored “foundation model*” and changed only the UML block from TITLE-ABS-KEY to TITLE-ABS OR AUTHKEY .	250	5/5	The field modification removed 126 baseline records while preserving all seeds. Differential inspection showed substantial removal of records whose UML association originated only from database index terms, while relevant or potentially eligible studies whose UML relationship appeared in author keywords were preserved. The field modification was accepted.

Round	Controlled modification	Scopus N	Seeds	Interpretation and decision
S4	Combined the two independently accepted modifications: removed “foundation model*” and searched the UML block in TITLE-ABS OR AUTHKEY.	249	5/5	S4 contained exactly the intersection of the unique-record sets retrieved by S2 and S3. Relative to S3, only the 2002 FMCO proceedings record was removed. No prespecified seed was lost. S4 was accepted as Scopus search-string version 1.0 for subsequent source validation and production searching.

The diagnostic unique-record counts were 372 for S1, 367 for S2, 246 for S3, and 245 for S4. S4 therefore removed 127 unique identities relative to the S1 baseline while preserving the complete prespecified seed set. These values are reported as validation diagnostics rather than as study-selection results.

1.8.3.3 Rationale for the accepted Scopus modifications

The S2 comparison showed that the generic term “foundation model*” was not sufficiently specific to the LLM construct in this review. Its exclusive retrieval contribution consisted of records outside the intended scope; therefore, the term was removed rather than retained solely to increase nominal recall.

The S3 comparison demonstrated that using the composite Scopus keyword field for the UML block introduced substantial retrieval through database-assigned index terms. Restricting this block to title, abstract, or author keywords preserves documents in which UML is explicitly part of the authors’ description of the study while reducing retrieval caused only by controlled indexing terms. A title-and-abstract-only restriction was not adopted because differential inspection identified potentially eligible records whose UML relationship was expressed in author keywords.

These decisions are specific to the Scopus field model. They will not be mechanically transferred to other databases. Each source-specific query will preserve the same conceptual blocks while using the source’s documented field semantics and will undergo its own seed and differential-record validation.

1.8.3.4 Validated Scopus search string v1.0

The following query was accepted after S4. No temporal, subject-area, language, or document-type filter was applied during the lexical and field validation so that field effects could be evaluated independently. Such restrictions, when required by the protocol, are applied and recorded separately during production searching.

```

TITLE-ABS-KEY (
  ( "large language model*" OR LLM* OR "generative AI" OR
    "generative artificial intelligence" OR ChatGPT OR GPT OR Claude OR
    Gemini OR Llama OR Qwen OR DeepSeek )
)
AND
(
  TITLE-ABS (
    UML OR "Unified Modeling Language" OR PlantUML OR
    "class diagram*" OR "use case diagram*" OR "sequence diagram*" OR
    "activity diagram*" OR "state machine diagram*" OR "state diagram*" OR
    "component diagram*" OR "deployment diagram*" OR "object diagram*" OR
    "communication diagram*" OR "package diagram*" OR
    "composite structure diagram*" OR "timing diagram*" OR
    "interaction overview diagram*"
  )
  OR
  AUTHKEY (
    UML OR "Unified Modeling Language" OR PlantUML OR
    "class diagram*" OR "use case diagram*" OR "sequence diagram*" OR
    "activity diagram*" OR "state machine diagram*" OR "state diagram*" OR
    "component diagram*" OR "deployment diagram*" OR "object diagram*" OR
    "communication diagram*" OR "package diagram*" OR
    "composite structure diagram*" OR "timing diagram*" OR
    "interaction overview diagram*"
  )
)
AND
TITLE-ABS-KEY (
  ( generat* OR creat* OR construct* OR synthes* OR produc* OR transform* OR
    complet* OR repair* OR refine* OR "natural language" OR requirement* OR
    specification* OR "user stor*" OR scenario* OR "problem statement*" OR
    "domain description*" )
)

```

The complete RIS exports for S1–S4 are preserved in controlled project storage for audit. Their redistribution in the public replication package occurs only when the source terms permit it. Independently of that redistribution, the public replication package preserves the query, validation log, counts, permitted audit metadata, and reproduction information. Source-specific validation of ACM Digital Library, IEEE Xplore, Scopus-independent indexing services, and the complementary sources remains subject to the same protocol before the complete multi-source strategy is frozen.

1.8.3.5 IEEE Xplore source-specific validation

IEEE Xplore was validated independently rather than receiving a mechanical translation of the Scopus field configuration. This decision follows the protocol principle that the conceptual search architecture must remain stable across sources while field semantics, indexing practices, syntax, and interface behavior are validated within each information source.

Invalid implementation test I0. An initial implementation using the IEEE Xplore Structured Advanced Search interface split the UML concept block across two boxes joined by OR, followed by the LLM and generation blocks joined by AND. Inspection of the Boolean expression generated by the interface showed that the intended grouping

$$(UML_A \vee UML_B) \wedge LLM \wedge GEN$$

was not preserved. Instead, the generated expression was equivalent to

$$UML_A \vee (UML_B \wedge LLM \wedge GEN),$$

which returned 33,888 records. Because the implementation did not represent the intended conceptual query, I0 was invalidated and is not treated as a content-sensitivity round. Subsequent IEEE Xplore validation used Command Search with explicit parentheses so that operator grouping remained auditable.

IEEE-specific seed denominator. The five prespecified cross-source seeds were individually searched by DOI in IEEE Xplore before query validation. De Bari et al. [2024] (DOI <https://doi.org/10.1145/3674805.3690741>) was empirically confirmed as not indexed in IEEE Xplore. The remaining four seeds were indexed and therefore constitute the IEEE-specific validation denominator

$$|S_{\text{IEEE}}| = 4.$$

They are Ferrari et al. [2024], Garaccione et al. [2025a], Garaccione et al. [2025b], and Xiao et al. [2025]. All four were retrieved in I1, I2, and I3, corresponding to $4/4 = 100\%$ seed recovery in every valid IEEE round.

IEEE Xplore validation rounds. I1 used a Command Search representation of the three conceptual blocks and searched every block in *All Metadata*. I2 changed only the UML block, restricting it to *Document Title*, *Abstract*, or *Author Keywords*. I3 restored the UML block to *All Metadata* and restricted only the LLM block to *Document Title*, *Abstract*, or *Author Keywords*. No temporal, language, subject-area, publication-title, or document-type filters were applied during these rounds.

Table 1.11: IEEE Xplore search-strategy validation log

Round	Controlled implementation or modification	IEEE N	Seeds	Interpretation and decision
I0	Structured Advanced Search implementation with the UML block split across two boxes joined by OR, followed by the remaining concept blocks.	33,888	n/a	Invalid implementation. The interface-generated Boolean expression did not preserve the intended grouping because AND bound more strongly than OR. The result was discarded as a query-implementation failure rather than evaluated as a search-strategy variant.
I1	Valid Command Search baseline. LLM, UML, and generation/input blocks searched in <i>All Metadata</i> .	327	4/4	Accepted as the valid baseline. Inspection showed precision loss from ambiguous LLM model names occurring in affiliations or unrelated metadata and substantial retrieval through broad UML metadata/indexing.

Round	Controlled implementation or modification	IEEE N	Seeds	Interpretation and decision
I2	Changed only the UML block from <i>All Metadata</i> to <i>Document Title OR Abstract OR Author Keywords</i> .	104	4/4	Rejected. Although all four indexed seeds were retained, differential inspection identified potentially relevant records removed by the restriction, including studies explicitly discussing class, sequence, activity, or other diagrams. The change also exposed lexical sensitivity to singular/plural diagram expressions once IEEE indexing fields were removed. I2 therefore did not preserve recall adequately.
I3	Restored the UML block to <i>All Metadata</i> and changed only the LLM block to <i>Document Title OR Abstract OR Author Keywords</i> .	283	4/4	Accepted as IEEE Xplore search-string version 1.0. I3 removed 44 I1 records while retaining all indexed seeds. Differential inspection indicated that the removed records predominantly reflected ambiguous occurrences of model names such as Claude or Gemini in affiliations or unrelated metadata. No clearly eligible recent study was identified among the removed set.

Interpretation of the IEEE field tests. The I2 and I3 results demonstrate that source-specific field behavior cannot be assumed to reproduce the Scopus behavior. In Scopus, restricting the UML block to title, abstract, or author keywords reduced index-term noise without observed loss of relevant studies in the differential set. In IEEE Xplore, the analogous restriction produced potential false negatives and was therefore rejected. Conversely, restricting the LLM block was beneficial in IEEE Xplore because terms such as *Claude* and *Gemini* can occur in affiliations, organization names, historical technology names, or other peripheral metadata. The accepted IEEE configuration therefore deliberately prioritizes sensitivity for the UML concept while increasing specificity for the LLM concept.

1.8.3.6 Validated IEEE Xplore search string v1.0

The accepted IEEE Xplore strategy uses Command Search with explicit Boolean grouping. The LLM block is restricted to *Document Title*, *Abstract*, or *Author Keywords*; the UML and generation/input blocks remain in *All Metadata*. The generic “foundation model*” term remains excluded from the LLM concept block according to the lexical decision established during the Scopus sensitivity analysis.

```

(
(
  "Document Title":"large language model*" OR
  "Document Title":LLM OR "Document Title":LLMs OR
  "Document Title":"generative AI" OR
  "Document Title":"generative artificial intelligence" OR
  "Document Title":ChatGPT OR "Document Title":GPT OR
  "Document Title":Claude OR "Document Title":Gemini OR
  "Document Title":Llama OR "Document Title":Qwen OR
  "Document Title":DeepSeek OR
  "Abstract":"large language model*" OR "Abstract":LLM OR
  "Abstract":LLMs OR "Abstract":"generative AI" OR
  "Abstract":"generative artificial intelligence" OR
  "Abstract":ChatGPT OR "Abstract":GPT OR "Abstract":Claude OR
  "Abstract":Gemini OR "Abstract":Llama OR "Abstract":Qwen OR
  "Abstract":DeepSeek OR
  "Author Keywords":"large language model*" OR
  "Author Keywords":LLM OR "Author Keywords":LLMs OR
  "Author Keywords":"generative AI" OR
  "Author Keywords":"generative artificial intelligence" OR
  "Author Keywords":ChatGPT OR "Author Keywords":GPT OR
  "Author Keywords":Claude OR "Author Keywords":Gemini OR
  "Author Keywords":Llama OR "Author Keywords":Qwen OR
  "Author Keywords":DeepSeek
)
AND
(
  "All Metadata":UML OR "All Metadata":"Unified Modeling Language" OR
  "All Metadata":PlantUML OR "All Metadata":"class diagram" OR
  "All Metadata":"use case diagram" OR "All Metadata":"sequence diagram" OR
  "All Metadata":"activity diagram" OR
  "All Metadata":"state machine diagram" OR "All Metadata":"state diagram" OR
  "All Metadata":"component diagram" OR "All Metadata":"deployment diagram" OR
  "All Metadata":"object diagram" OR "All Metadata":"communication diagram" OR
  "All Metadata":"package diagram" OR
  "All Metadata":"composite structure diagram" OR
  "All Metadata":"timing diagram" OR
  "All Metadata":"interaction overview diagram"
)
AND
(
  "All Metadata":generate OR "All Metadata":generation OR
  "All Metadata":create OR "All Metadata":creation OR
  "All Metadata":construct OR "All Metadata":construction OR
  "All Metadata":synthesis OR "All Metadata":produce OR
  "All Metadata":transform OR "All Metadata":transformation OR
  "All Metadata":complete OR "All Metadata":completion OR
  "All Metadata":repair OR "All Metadata":refine OR
  "All Metadata":refinement OR "All Metadata":"natural language" OR
  "All Metadata":requirement OR "All Metadata":specification OR
  "All Metadata":"user story" OR "All Metadata":scenario OR
  "All Metadata":"problem statement" OR
  "All Metadata":"domain description"
)
)
)

```

The I0 implementation record and the I1–I3 raw CSV exports are preserved in controlled project storage. Raw exports are redistributed publicly only when permitted. The public replication package preserves the validation log, query, counts, decisions, and permitted audit metadata. The production IEEE Xplore retrieval will apply the protocol’s temporal and other explicitly approved eligibility restrictions only after this field-level query validation, thereby preserving the separation between lexical/field validation and production filtering.

1.8.3.7 ACM Digital Library source-specific validation

The ACM Digital Library was validated using the *ACM Full-Text Collection* rather than the broader ACM Guide to Computing Literature. The Full-Text Collection was selected so that ACM Digital Library acts in the review as a direct source of ACM-published literature rather than as a second multidisciplinary bibliographic index overlapping extensively with Scopus and

Web of Science.

Interface implementation checks. Validation used the Advanced Search interface available through the institutional CAPES proxy. The interface exposes *Anywhere*, *Title*, *Abstract*, *Full Text*, *Author Affiliation*, *Author Keyword*, DOI, and additional bibliographic fields. Empirical interface testing established three implementation properties that affect reproducibility.

- 1) Distinct *Search Within* rows are combined by AND and the interface does not expose a raw Boolean editor that permits changing the operator between rows.
- 2) Field prefixes typed manually inside one *Anywhere* row are parsed and preserve explicit OR relations across fields.
- 3) The manually typed prefix **Keyword:** was operationally equivalent to selecting the visual *Author Keyword* field in the interface during a dedicated equivalence check.

Consequently, field-restricted disjunctions were implemented inside a single *Anywhere* row, for example

$$Title(LLM) \vee Abstract(LLM) \vee AuthorKeyword(LLM),$$

while the three conceptual blocks remained connected by the interface-generated AND relation between rows.

ACM-specific seed denominator. The five prespecified cross-source seeds were searched individually by DOI in the ACM Full-Text Collection. Only De Bari et al. [2024] was indexed in this collection. Ferrari et al., Garaccione et al. (class diagrams), Garaccione et al. (use case diagrams), and Xiao et al. were not retrieved by DOI from the ACM Full-Text Collection and therefore do not belong to the source-specific validation denominator. The ACM denominator is consequently

$$|S_{ACM}| = 1.$$

De Bari et al. was retrieved by A1, A2, and A3, corresponding to $1/1 = 100\%$ seed recovery in every executed ACM validation round. Because this denominator contains only one seed, source validation relied primarily on differential-record inspection and diagnostic sensitivity cases rather than seed recovery alone.

ACM validation rounds. A1 used three *Anywhere* rows representing the LLM, UML, and generation/input concept blocks. A2 changed only the LLM block, requiring an LLM term to occur in the title, abstract, or author keywords, while UML and generation/input remained in *Anywhere*. A3 restored the LLM block to *Anywhere* and changed only the UML block to title, abstract, or author keywords. A4, which would have combined the A2 and A3 modifications, was not executed because A3 demonstrated a relevant false-negative risk that A4 would necessarily inherit.

Table 1.12: ACM Full-Text Collection search-strategy validation log

Round	Controlled modification	ACM N	Seed	Interpretation and decision
A1	Baseline. LLM, UML, and generation/input blocks searched through three <i>Anywhere</i> rows in the ACM Full-Text Collection.	1,854	1/1	Valid broad baseline but rejected as a production configuration because inspection showed substantial retrieval of pre-LLM and topically unrelated literature attributable to broad-field matching. The available EndNote export contained 1,000 records and was retained as a partial diagnostic export rather than represented as the complete A1 retrieval set.
A2	Changed only the LLM block to <i>Title</i> $\vee$ <i>Abstract</i> $\vee$ <i>AuthorKeyword</i> ; UML and generation/input remained in <i>Anywhere</i> .	454	1/1	Accepted. The modification reduced the broad baseline substantially, concentrated retrieval in contemporary LLM literature, preserved the indexed seed, and did not reveal a clearly eligible false negative in the observable differential evidence available from A1.
A3	Restored LLM to <i>Anywhere</i> and changed only the UML block to <i>Title</i> $\vee$ <i>Abstract</i> $\vee$ <i>AuthorKeyword</i> .	104	1/1	Rejected. Differential inspection identified relevant or potentially eligible model-generation studies that did not expose UML terminology in title, abstract, or author keywords. A diagnostic false-negative case was <i>Multi-step Iterative Automated Domain Modeling with Large Language Models</i> Yang et al. [2024], which uses natural-language problem descriptions to generate and evaluate class diagrams but would be lost under the A3 field restriction.

Round	Controlled modification	ACM N	Seed	Interpretation and decision
A4	Planned combination of the independently tested A2 and A3 field restrictions.	not executed	n/a	Cancelled. Because A3 had already demonstrated a relevant false-negative risk, any intersection inheriting the A3 UML-field restriction would preserve that loss and would not provide a defensible recall-preserving refinement.

Interpretation of the ACM field tests. The ACM validation demonstrates a field behavior different from both Scopus and IEEE Xplore. Restricting the LLM block to title, abstract, or author keywords was beneficial because it reduced matches in which LLM terminology was only incidental in the full text or other broad searchable content. Restricting the UML block, however, was not acceptable because substantively relevant MDE papers may use broader expressions such as *domain modeling*, *model generation*, classes, attributes, and relationships in their title or abstract while specifying UML class diagrams only in the body of the report. The accepted ACM configuration therefore increases specificity for the LLM construct while deliberately preserving a broad UML search field.

The A3 false-negative evidence is retained as a diagnostic sensitivity case and is not retroactively added to the prespecified seed set. This distinction preserves the prospective role of the formal seed set while allowing studies discovered during validation to inform field-sensitivity decisions.

1.8.3.8 Validated ACM Full-Text Collection search strategy v1.0

The accepted ACM configuration is A2:

$$LLM_{Title \vee Abstract \vee AuthorKeyword} \wedge UML_{Anywhere} \wedge Generation_{Anywhere}.$$

The query is implemented using three *Search Within* rows. The first row is set to *Anywhere* but contains explicit field prefixes so that the LLM block is evaluated across title, abstract, or author keywords. The second and third rows use *Anywhere*; the interface combines the three rows using AND.

Row 1: field-restricted LLM block.

```
Title:(
  "large language model" OR "large language models" OR LLM OR LLMs OR
  "generative AI" OR "generative artificial intelligence" OR ChatGPT OR GPT OR
  Claude OR Gemini OR Llama OR Qwen OR DeepSeek
)
OR Abstract:(
  "large language model" OR "large language models" OR LLM OR LLMs OR
  "generative AI" OR "generative artificial intelligence" OR ChatGPT OR GPT OR
  Claude OR Gemini OR Llama OR Qwen OR DeepSeek
)
OR Keyword:(
  "large language model" OR "large language models" OR LLM OR LLMs OR
  "generative AI" OR "generative artificial intelligence" OR ChatGPT OR GPT OR
  Claude OR Gemini OR Llama OR Qwen OR DeepSeek
)
```

Row 2: broad UML block.

UML OR "Unified Modeling Language" OR PlantUML OR
"class diagram" OR "class diagrams" OR "use case diagram" OR "use case diagrams" OR
"sequence diagram" OR "sequence diagrams" OR "activity diagram" OR "activity diagrams" OR
"state machine diagram" OR "state machine diagrams" OR "state diagram" OR "state diagrams" OR
"component diagram" OR "component diagrams" OR "deployment diagram" OR "deployment diagrams" OR
"object diagram" OR "object diagrams" OR "communication diagram" OR "communication diagrams" OR
"package diagram" OR "package diagrams" OR
"composite structure diagram" OR "composite structure diagrams" OR
"timing diagram" OR "timing diagrams" OR
"interaction overview diagram" OR "interaction overview diagrams"

Row 3: broad generation/input block.

generate OR generated OR generating OR generation OR
create OR created OR creating OR creation OR
construct OR constructed OR construction OR
synthesize OR synthesized OR synthesis OR produce OR produced OR
transform OR transformed OR transformation OR complete OR completion OR repair OR
refine OR refinement OR "natural language" OR requirement OR requirements OR
specification OR specifications OR "user story" OR "user stories" OR
scenario OR scenarios OR "problem statement" OR "problem statements" OR
"domain description" OR "domain descriptions"

No temporal, language, publication-type, or venue filter was used during A1–A3 field validation. Such restrictions are applied and documented separately during production retrieval. The complete A2 and A3 EndNote exports and the partial A1 diagnostic export are preserved in controlled project storage. .enw files are redistributed in the public replication package only when the applicable terms permit it. The seed-indexing check, field-prefix equivalence check, validation log, counts, diagnostic decisions, and permitted audit metadata remain documented in the replication package.

1.8.4 Complementary search

Backward and forward snowballing will be conducted according to a predefined and recorded procedure Wohlin [2014]. The start set comprises all included studies and relevant secondary studies. Backward snowballing inspects references. Forward snowballing uses Google Scholar and, when available, Scopus or Web of Science citation links. Newly identified records are deduplicated and screened with the same criteria used for database results. Iterations continue until one complete iteration produces no new eligible primary study.

The automated search will be updated immediately before the final synthesis and reporting. The update will use the same sources, concepts, and eligibility criteria. Changes caused by database syntax or platform evolution will be documented.

1.9 Eligibility Criteria

1.9.1 Screening gate structure

Amendment A004 restructures screening so that every decision point carries exactly one question, at most one inclusion criterion, and at most one exclusion criterion. The criteria tables that follow are read against this structure: the code records the identity of the criterion, and the gate outcome field records where it was applied.

Table 1.13: Screening gate structure after amendment A004

Gate	Filter	Question	Inclusion	Exclusion
—	D	Is the record a duplicate or a less complete member of a publication family?	—	E3
A	A1	Is the report within the temporal and language scope?	I1	E4

Gate	Filter	Question	Inclusion	Exclusion
A	A2	Is the report a complete scientific report?	I2	E1
A	A3	Is the report a primary study?	I3	E2
B	B0	Does the LLM generate or alter the UML content?	I4, I5, I7	—
B	B1	Does the generated result include UML content?	I7	E7
B	B2	Is the UML content produced or altered?	I5	E8
B	B3	Is the source input textual?	I6	E9
B	B4	What is the operational role of the model?	I4	E6
B	B5	Is the UML content separable from other notations?	I8	E7b
C	C1	Is there an identifiable generation instance?	I8	E12

Filter D is a corpus-level pre-pass reported as *records removed before screening*, consistent with PRISMA 2020. Filter A3 routes secondary and tertiary reports to a materialized background pile rather than discarding them. Filter B0 fast-paths clear cases, but a negative answer at B0 does not by itself exclude: it forces descent to B1 through B4 so that the recorded code always originates from a named filter. Filter C1 is the only exclusion available at Gate C.

Filters B1 through B4 are additionally recorded as descriptive attributes even when B0 absolves, without power to exclude. This front-loads three of the review’s incommensurability dimensions, namely diagram type, source specification, and model family, at no additional reading cost. Two rules accompany the practice: screening attributes are provisional and are stored in fields separate from extraction fields, and attributes do not enter the inter-rater agreement calculation, which covers only B0 and the named criterion.

1.9.2 Inclusion criteria

Inclusion codes are numbered in filter order rather than historical order. Each inclusion criterion is the positive counterpart of exactly one exclusion criterion, with the two exceptions stated after the table.

Table 1.14: Inclusion criteria

Code	Criterion	Filter	Pairs with
I1	The report is within the temporal and language scope of the review.	A1	E4
I2	The report is a complete scientific report.	A2	E1
I3	The report is a primary study.	A3	E2
I4	The study uses at least one LLM as a substantive component of the diagram generation process.	B4	E6
I5	The task produces, transforms, completes, repairs, refines, or revises UML diagram content.	B2	E8
I6	The source input includes textual software requirements, user stories, scenarios, specifications, problem statements, or explicit textual domain descriptions.	B3	E9
I7	The generated result includes explicit UML diagram content, such as PlantUML code, another textual UML representation, XMI, a rendered UML diagram, or another representation from which UML content can be assessed.	B1	E7

Code	Criterion	Filter	Pairs with
I8	The UML diagram content is separable, both from other notations and from other generated artifacts, tasks, or outputs.	B5, C1	E7b, E12

Two structural asymmetries are deliberate. E3 has no positive counterpart, because it predicates a property of the record set rather than of the report. Filter B0 has no criterion of its own, being the conjunction of I4, I5, and I7; it absolves but does not condemn.

I8 is evaluated at two moments without receiving a separate code for each: at B5 from title and abstract, against mixture with another notation, and at C1 from the full text, against mixture with other artifacts, tasks, or outputs. The moment of evaluation is recorded in the gate outcome field. Creating one code per reading moment would enlarge the table without adding discriminating power.

1.9.3 Exclusion criteria

Exclusion codes are frozen. Codes withdrawn by amendment A004 leave the operative table and remain recorded as declared gaps, so that the numbering of decisions taken before the amendment is not rewritten and the trace of the amendment is not erased.

Table 1.15: Exclusion criteria

Code	Criterion	Filter
E1	The item is not a complete scientific report, such as an editorial, preface, keynote, tutorial, slide deck, poster-only item, abstract-only record, thesis, book, patent, or non-scientific web content. This list is closed.	A2
E2	The report is a secondary or tertiary study.	A3
E3	The report is a duplicate or a less complete member of a publication family.	D
E4	The report is outside the temporal or language scope of the review.	A1
E6	The study does not use an LLM as a substantive component in producing, transforming, completing, repairing, refining, or revising UML diagram content.	B4
E7	The generated result is not UML. This includes generic diagram generation and the generation of code, metamodels, or other non-UML artifacts from a UML input.	B1
E7b	The generated result contains UML mixed with another notation, such as C4, ER, BPMN, SysML, Mermaid, or an architecture sketch, with no separable UML contribution.	B5
E8	The task only evaluates, explains, summarizes, classifies, or discusses an existing UML diagram, without producing, transforming, completing, repairing, refining, or revising UML diagram content.	B2
E9	The source input is code, an image, an existing UML model, logs, or another non-textual artifact without a substantive textual requirements or domain description component.	B3

Code	Criterion	Filter
E12	There is no identifiable generation instance: the UML result cannot be separated from other artifacts, tasks, or outputs generated in the same execution, so that no reported result is attributable to the generation of UML content.	C1

Table 1.16: Exclusion criteria withdrawn by amendment A004, kept as declared gaps

Code	Disposition
E5	Withdrawn. Full-text unavailability is attrition, not ineligibility, and is handled by the stratum of Section 1.9.6.
E10	Withdrawn. Split into filter C1 with code E12 and the extraction attribute <i>result attribution</i> .
E11	Withdrawn. Replaced by the three extraction axes of Section 1.9.5. Extractable quality evidence is no longer an eligibility requirement.

Bare E7 always denotes the first clause of the criterion as originally published, namely that the generated result is not UML. The sibling clause carries the suffixed code E7b, whose suffix preserves its provenance. E12 receives a new number rather than the vacated E10 because its statement changed from a negative aggregate to a positive verifiable question, and a new statement receives a new code.

When more than one exclusion criterion applies, the first criterion that clearly explains the exclusion in filter order is recorded as primary, and the remaining ones are recorded in the justification. If title and abstract are insufficient to decide whether the report involves LLM-based UML generation from textual requirements or domain descriptions, the report is retained for full-text screening. The absence of explicit syntactic or semantic terminology is never sufficient for exclusion, at any gate, because the review’s search was deliberately designed without quality terms and selecting on evaluation vocabulary would amount to selecting on the review’s own dependent variable.

1.9.4 Operational boundary for substantive LLM use

A report satisfies the LLM criterion only when the language model is used as a substantive component in producing, transforming, completing, repairing, refining, or revising the generated UML diagram content. Substantive use means that the model contributes directly to domain-bearing UML content, such as classes, attributes, operations, actors, use cases, relations, messages, lifelines, states, transitions, activities, components, dependencies, or other UML elements and relations.

Model names or broad labels are not sufficient for inclusion. Terms such as pretrained language model, foundation model, or language model support inclusion only when the operational role of the model satisfies the substantive-use requirement. Encoder-only or representation models, including BERT-like models, do not satisfy the LLM criterion solely because they are pretrained language models. When such models are used only as classifiers, sequence taggers, embedding generators, named-entity recognizers, relation extractors, or supervised preprocessing components, the report is excluded under E6.

Hybrid neural-symbolic pipelines are eligible when the LLM determines, proposes, or revises the semantic content of the UML diagram and symbolic components format, serialize, validate, render, or repair the representation. They are not eligible when symbolic rules determine the

UML content and the language model only performs peripheral preprocessing, summarization, formatting, evaluation, or explanation.

When title and abstract do not make the role of the model clear, the report is retained for full-text screening. The exclusion decision is made only when the full text provides sufficient evidence that the language model does not have a substantive role in producing, transforming, completing, repairing, refining, or revising UML diagram content.

1.9.5 Quality evidence classification during extraction

The review does not define a second eligibility layer, and evidence of quality is not an eligibility filter. All included studies satisfy the same inclusion and exclusion criteria. During data extraction, each included study is classified according to the type and strength of quality evidence that it provides for the generated UML diagram content. These classifications support synthesis without redefining the eligible corpus.

The classification is recorded on three independent axes derived from the operational definitions of Section 1.6 and from the model quality relations of Lindland et al. and Krogstie Lindland et al. [1994], Krogstie et al. [2006].

Table 1.17: Quality evidence axes recorded during extraction

Axis	Name	Relation	Values
L	Language	Model to language	Absent; textual validity; UML conformity.
D	Domain	Model to domain	Absent; claimed only; source requirements; reference model; expert judgment; rubric.
U	Use	Model to interpreter and use	Absent; claimed only; comprehension; engineering activity; rework.

The three axes are never collapsed into a single scale. Folding pragmatic adequacy into the domain axis would repeat the failure that produced an overloaded exclusion code during the first screening pass: a fused question yields a fused code, and the distinction becomes unrecoverable.

A fourth field, inherited from the withdrawn E10, records *result attribution* with values *attributable to UML*, *aggregated with other artifacts*, and *not reported*. It qualifies the strength with which a reported result can be traced to the generation of UML content, in cases where filter C1 confirmed that a generation instance exists.

The extraction additionally classifies which quality constructs the study provides evidence about. That vocabulary is defined once, in the normalized construct field of the quality evidence extraction table, whose syntactic and semantic projections are the two construct fields recorded beside it; it is deliberately not repeated here, because repeating it is what allowed the two enumerations to drift apart. The extraction also records whether syntactic and semantic evidence are reported separately and whether the report provides direct, indirect, absent, unclear, or non-assessable evidence of syntax-semantic dissonance. Axis L is the study-level projection of the syntactic construct field onto the two levels the glossary separates. *Textual validity*, *rendering validity*, *PlantUML parseability*, and *notation-level validity* project onto the textual level, because textual validity is defined as conformance of the concrete output string to the grammar accepted by the selected representation or rendering tool. *UML syntactic conformity*, *metamodel conformance*, and *well-formedness* project onto the UML level, because UML syntactic conformity is defined over the abstract syntax and its well-formedness constraints. No construct label is left without a level, so a report whose only syntactic evidence is a renderer run or a parser run has a value on axis L and is not recorded as absent. Because the axis carries one value per study while the construct field is repeatable, the axis records the most demanding

level for which the study reports evidence; this asserts which level was assessed and not that one level implies the other, and the enumeration of what was assessed remains in the construct field. Procedure, reliability, and validity information is recorded in the SQ5 extraction fields and never in the axes.

The analytical subsets are computed from these axes rather than presumed by an eligibility filter.

Table 1.18: Analytical subsets derived from the quality evidence axes

Questions	Subset
MQ1 to MQ5, and SQ6	The whole eligible corpus.
SQ1 to SQ3	The whole eligible corpus.
SQ4	Studies for which axis L is not absent and axis D is not absent.
SQ7	Studies for which axis U is not absent.

SQ4 asks to what extent the reported evidence distinguishes syntactic from semantic quality. Under an eligibility filter on evidence reporting, that subset would be assumed; under the axes it is computed, which is the difference between describing the literature and describing the review’s own filter. The same argument governs MQ5, which asks which combinations remain under-investigated: a gap analysis run over a corpus filtered by evidence reporting would report the filter as a gap in the literature. SQ6 covers the whole corpus for a related reason. What knowledge is supplied as generation context is a property of every generation setting, whereas the pragmatic effects asked by SQ7 exist only where axis U is not absent; reporting the former over the latter subset would describe the supply of domain knowledge and of the UML technical contract over the minority of studies that also measured use. SQ1 to SQ3 cover the whole corpus for the strongest form of the same reason. A quality construct, an inadequacy, and a metric may be purely syntactic: a study whose only evidence is a parser run reports a construct, a definition, and an oracle, and would be silently removed from the construct matrix by a subset conditioned on the domain axis. Restricting the question that carries the constitutive function of this review to the studies that already evaluate against the domain would select on the dependent variable and would report the review’s own filter as the vocabulary of the literature. The two rows above are kept separate rather than merged so that the line references recorded in the amendments and in the decision log remain valid.

1.9.6 Full-text unavailability as attrition

A report whose full text is not obtained after the documented access procedure is not excluded. It is placed in a stratum named *identified, not retrieved*. It receives neither an exclusion flag nor an exclusion code, and its gate C outcome is recorded as not retrieved.

Every eligibility criterion predicates something of the report: its year, its type, its object, the operational role of the model, the separability of its result. Full-text unavailability predicates something of the reviewer’s access within a time window. Reporting it as an exclusion would read as a statement about the ineligibility of those reports, when the correct statement is that they are reports that could not be read. PRISMA 2020 already separates the two into disjoint boxes, in which *reports not retrieved* precedes and does not overlap *reports excluded, with reasons* Page et al. [2021].

The decision is also empirical. Retrieval loss in this review is systematic rather than random along three dimensions, namely access model, publication type and source, and recency. An exclusion code would hide all three asymmetries, and the recency bias is the most damaging, because the object of the review is a fast-moving literature whose most recent slice is the most informative about model families and prompting strategies.

Reports in the stratum are characterized rather than merely counted, using the metadata and abstracts available for them. The cut-off date for access attempts is unchanged, but its effect changes from excluding reports to closing the stratum: on that date the stratum stops receiving new texts and is counted and described. Question-level bases are stated explicitly, distinguishing the descriptive questions answerable over all retained reports from the questions that require full text, with a sensitivity check of the former against the abstracts of the non-retrieved stratum.

1.9.7 Ambiguity rules

At title and abstract screening, uncertainty favors retention. A record is excluded only when an exclusion criterion is clearly satisfied. At full text screening, each decision must cite the relevant criterion and a brief justification. Multi artifact and multi task studies are included when UML specific data can be extracted. Educational studies are eligible because they provide empirical evidence about generation and assessment, but educational context is coded explicitly and considered during interpretation.

1.9.8 Publication families

Reports are grouped into publication families when they share a dataset, artifact, experimental design, tool, or substantial body of results. The most complete report serves as the primary extraction source. Complementary reports are used to fill missing details but are not counted as independent primary studies. Differences between family members are recorded, including changes in models, datasets, methods, or results.

1.10 Record Management and Deduplication

All database exports are preserved in their original formats before transformation. Each record receives a stable review identifier. Automated deduplication uses DOI, title, authors, year, and source identifiers, followed by manual inspection of probable duplicates. Preprint and published versions are linked rather than deleted before publication family assessment. The master record set preserves source provenance so that retrieval coverage by database can be analyzed.

The record management log will contain the following fields.

- stable record identifier;
- database or discovery source;
- database specific identifier;
- title, authors, year, venue, DOI, and URL;
- duplicate group, publication family identifier, and pre-pass D outcome;
- screening decisions by reviewer and stage, recorded per gate as an outcome that names the filter that decided the case;
- screening attributes annotated at filters B1 to B4, stored separately from extraction fields and excluded from the agreement calculation;
- uncertainty flags raised at title and abstract, which retain the record and never exclude it;
- exclusion criterion and justification, with one primary code per excluded record;
- full-text retrieval status, including membership of the *identified*, *not retrieved* stratum;
- eligibility decision and quality-evidence classification on axes L, D, and U;

- extraction and appraisal status;
- amendment or recoding flags.

1.11 Study Selection Process

Selection comprises calibration, deduplication as a corpus-level pre-pass, title and abstract screening, full text screening, quality-evidence classification, and publication family reconciliation.

1.11.1 Reviewer calibration

Before production screening, two reviewers independently evaluate a purposive sample containing clearly eligible, clearly ineligible, and ambiguous records. The sample must represent different UML diagram types and publication formats. Disagreements are discussed and the eligibility manual is revised. Calibration continues until the reviewers achieve at least 80 percent agreement and a Cohen kappa of at least 0.70 for the pilot set Cohen [1960]. The threshold is a procedural target rather than a claim that kappa alone demonstrates validity.

1.11.2 Title and abstract screening

The primary reviewer screens all deduplicated records. A second reviewer independently screens a stratified random sample of at least 20 percent and every record marked uncertain by the primary reviewer. If disagreement exceeds 10 percent or kappa is below 0.70, an additional 20 percent sample is independently screened. This expansion continues until the predefined threshold is met or all records have been dual screened.

1.11.3 Full text screening

Every record retained after title and abstract screening is assessed independently by two reviewers. Each reviewer records one of three decisions, include, exclude, or unclear. Excluded reports receive one primary exclusion criterion and a textual justification. Disagreements are resolved through discussion. Unresolved cases are decided by an adjudicator with expertise in Software Engineering and UML or LLM based modeling.

1.11.4 Quality-evidence classification

Extractable quality evidence is not assessed for eligibility at any gate. For included reports, the type and strength of syntactic, semantic, pragmatic, and correction-related evidence are classified during extraction on the three axes of Section 1.9.5, without creating an additional eligibility status and without affecting the composition of the eligible corpus. A report that evaluates the generated UML content without using the review’s terminology is classified, not excluded.

1.11.5 Selection reporting

The final report will present a PRISMA flow diagram and a table of full text exclusions. Counts will distinguish database search, manual search, citation searching, duplicates, non research items, included studies, extraction-derived quality-evidence groupings, and unique primary studies after publication family merging Page et al. [2021].

Two reporting boxes are kept disjoint, in accordance with Section 1.9.6. *Records removed before screening* carries the corpus-level pre-pass D. *Reports not retrieved* carries the attrition stratum and is reported with its size and with the distribution of its access model, publication

type, source, and year, so that the systematic character of the loss is visible rather than absorbed into an exclusion count. *Reports excluded, with reasons* carries only the eligibility criteria of Section 1.9, each with one primary code and the filter at which it was applied.

1.12 Data Extraction

Data extraction is guided directly by the research questions. Two linked sections of one structured extraction form are applied to every included primary study: mapping and generation-context fields, and quality-evidence fields. Each field has a definition, allowed values, examples, missing data code, and relationship to one or more research questions. Analytical subsets are derived from these fields and do not alter eligibility.

1.12.1 Mapping and generation-context extraction form

Table 1.19: Mapping and generation-context extraction framework

Facet	Field	Extraction rule
Bibliographic	Year, venue, publication type, status	Preserve published and online first dates; code peer review and preprint status separately
Bibliographic	Authors and affiliations	Extract organizations, countries, and academic or industrial participation
Study	Goal and research questions	Preserve verbatim formulation and code study purpose
Study	Research method	Classify experiment, quasi experiment, benchmark, case study, user study, demonstration, design and evaluation, qualitative study, or other
Study	Contribution type	Tool, method, model, framework, dataset, benchmark, empirical findings, taxonomy, or other
Artifact	DiagramType	Class diagram; use case diagram; sequence diagram; activity diagram; state machine diagram; state diagram; component diagram; deployment diagram; object diagram; communication diagram; package diagram; composite structure diagram; timing diagram; interaction overview diagram; multiple UML types; unclear.
Artifact	GenerationTask	Generation; transformation; completion; repair; refinement; iterative revision; mixed task; unclear.
Input	InputRequirementForm	Requirements document; user stories; use case narratives; scenarios; problem statements; domain descriptions; structured specifications; mixed textual input; unclear.
Input	Input structure and quality	Free text, template, semi structured, structured; reported ambiguity, inconsistency, incompleteness, or preprocessing

Facet	Field	Extraction rule
Domain data	Application domain and dataset	Domain, public or private status, size, provenance, reference availability
LLM	ModelFamily	Exact model family or model name reported by the study, such as GPT, ChatGPT, Claude, Gemini, Llama, Qwen, DeepSeek, T5, BART, BERT, RoBERTa, or other.
LLM	ModelArchitectureOrAccessMode	Autoregressive; encoder-decoder; encoder-only; conversational/instructional; API-based; open-weight; fine-tuned; unclear; not reported.
LLM	GenerativeOrInstructionalUse	Yes; No; Unclear.
LLM	LLMRoleInPipeline	Semantic content generator; semantic content reviser; candidate proposer; surface formatter; syntax repairer; validator; evaluator only; pre-processor only; embedding or classifier only; unclear.
LLM	SemanticContentAuthority	LLM-determined; hybrid LLM-symbolic; symbolic/rule-determined; unclear.
LLM	SymbolicPipelineRole	None reported; formatting; serialization; validation; rendering; syntax repair; semantic rule generation; content generation; unclear.
LLM	SubstantiveLLMUseDecision	Included as substantive LLM use; excluded as non-substantive language-model use; unclear at title and abstract; resolved at full text.
LLM	BoundaryDecisionRationale	Short textual explanation of how the reviewers decided whether the model had a substantive role in producing, transforming, completing, repairing, refining, or revising UML diagram content.
Context	DomainKnowledgeProvided	None reported; implicit in requirements; explicit domain description; examples; ontology or structured knowledge; retrieval augmented context; expert-provided context; unclear.
Context	UMLTechnicalInstructionProvided	None reported; notation instruction; meta-model or well-formedness instruction; PlantUML syntax instruction; examples; tool or parser feedback; unclear.
Context	PromptingOrContextStrategy	Zero-shot; few-shot; chain-of-thought or reasoning instruction; retrieval augmented generation; iterative prompting; agentic workflow; fine tuning; tool feedback loop; mixed; unclear.
Context	Adaptation and orchestration	Fine tuning, preference optimization, agents, tools, self correction, iterative feedback, or none
Inference	Sampling and repetition	Temperature, top p, top k, seed, number of generations, selection policy

Facet	Field	Extraction rule
Output	OutputRepresentation	PlantUML code; rendered UML diagram; XMI; textual UML description; image; mixed representation; other explicit UML representation; unclear.
Output Evaluation	PlantUMLGenerated BaselineCondition	Yes; No; Unclear. No baseline; human-authored model; rule based or deterministic tool; another LLM; another version or configuration of the same LLM; ablation of the generation context; published benchmark or gold standard; other. Repeatable. Reported quality dimensions are not recorded here: they are given by the three axes and by the dimension partition of the normalized construct field.
Evaluation	Human involvement	Author inspection, domain expert, UML expert, student, practitioner, independent evaluator, or participant
Open science	Data, prompts, code, models, and replication package	Record availability, persistent identifier, access restrictions, and completeness

1.12.2 Quality-evidence extraction form

Table 1.20: Quality-evidence extraction framework

Facet	Field	Extraction rule
Axis	QualityAxisL	Absent; textual validity; UML conformity. Records the evidence reported about the relation between the model and its language.
Axis	QualityAxisD	Absent; claimed only; source requirements; reference model; expert judgment; rubric. Records the evidence reported about the relation between the model and the relevant domain.
Axis	QualityAxisU	Absent; claimed only; comprehension; engineering activity; rework. Records the evidence reported about the relation between the model and its interpreters or uses.
Axis	ResultAttribution	Attributable to UML; aggregated with other artifacts; not reported. Qualifies how far a reported result can be traced to the generation of UML content. Inherited from the withdrawn E10.
Construct	Original quality term	Preserve the term exactly as reported, before any mapping to the review’s scheme
Construct	Definition and theoretical source	Extract verbatim or near verbatim definition within copyright limits; record cited framework

Facet	Field	Extraction rule
Construct	Normalized construct	Single normalization vocabulary, partitioned by dimension. Syntactic: textual validity; rendering validity; PlantUML parseability; UML syntactic conformity; metamodel conformance; well-formedness; notation-level validity. Semantic: semantic correctness; semantic completeness; requirements coverage; domain fidelity; consistency; consistency with source requirements; omitted element; unsupported addition; incorrect relation; incorrect actor; incorrect behavior. Pragmatic: pragmatic adequacy; understandability; readability; clarity; correction effort; inspection effort. Or other. The two construct fields below are the syntactic and semantic projections of this vocabulary and admit no value absent from it.
Construct	SyntacticEvidenceAvailable	Yes; No; Unclear.
Construct	SemanticEvidenceAvailable	Yes; No; Unclear.
Construct	SyntacticQualityConstruct	Textual validity; rendering validity; PlantUML parseability; UML syntactic conformity; metamodel conformance; well-formedness; notation-level validity; other; not reported.
Construct	SemanticQualityConstruct	Semantic correctness; semantic completeness; requirements coverage; domain fidelity; consistency with source requirements; omitted element; unsupported addition; incorrect relation; incorrect actor; incorrect behavior; other; not reported.
Metric	SyntacticMetricOrProcedure	Exact metric, rubric, parser, compiler, rendering check, metamodel check, checklist, human judgment, automated tool, or narrative procedure reported by the study.
Metric	SemanticMetricOrProcedure	Exact metric, rubric, oracle, reference model, gold standard, requirements traceability procedure, expert judgment, checklist, human coding, automated comparison, or narrative procedure reported by the study.
Reference	EvaluationReference	Requirements text; domain description; reference UML model; gold standard; rubric; expert judgment; benchmark; parser or renderer; UML specification; metamodel; other; unclear.
Reference	OracleOrGoldStandard	Yes; No; Partial; Unclear.
Reference	Reference construction and validation	Who created it, from which sources, through which process, and whether disagreement or alternative valid models were handled
Unit	Unit of assessment	Diagram, requirement, proposition, actor, use case, class, attribute, relation, message, state, transition, or other

Facet	Field	Extraction rule
Inadequacy	Original label and definition	Preserve the source label and its operational meaning
Inadequacy	Normalized discrepancy operation	Omission, unsupported addition, incorrect representation, incorrect classification, inconsistency, duplication, redundancy, or emergent category. Selecting <i>emergent category</i> requires recording the emergent label in free text
Inadequacy	Violated reference	Concrete representation, UML abstract syntax, UML construct semantics, domain or requirements semantics, cross diagram consistency, or intended engineering task
Inadequacy	UML carrier	Affected diagram, element, relation, property, constraint, sequence, state, transition, message, behavior, or other type specific unit
Metric	Metric name and formula	Extract formula, numerator, denominator, scale, direction, threshold, and aggregation rule
Metric	Automation level	Fully automated, tool assisted, human coded, LLM assessed, or mixed
Evaluation	HumanEvaluatorRole	Author; independent evaluator; domain expert; UML expert; student; practitioner; crowd worker; not reported; not applicable.
Evaluation	AutomatedEvaluatorTool	PlantUML; parser; renderer; model checker; custom script; LLM-as-judge; other; none reported.
Evaluation	QualityAggregationRule	Per diagram; per element; per relation; per requirement; per prompt; per model; percentage; score; ordinal scale; pass/fail; narrative only; unclear.
Reliability	Agreement and uncertainty	Agreement measure, value, confidence interval, repeated generation, variance, effect size, significance, or qualitative confidence
Context relation	Context factor	Factor manipulated, controlled, observed, or merely reported
Context relation	Inferential status	Descriptive, comparative, relational, causal, mechanistic, or unsupported claim
Pragmatic use	Task, actor, and outcome	Engineering activity, participant profile, success criterion, time, accuracy, decision quality, or comprehension
Rework	CorrectionOrReworkEvidence	Yes; No; Unclear.
Rework	Correction measure	Time, edit count, changed elements, distance, severity, number of iterations, subjective effort, or other
Syntax-semantic dissonance	SyntaxSemanticDissonanceEvidence	Direct; Indirect; Absent; Unclear; Not assessable.

Facet	Field	Extraction rule
Syntax-semantic dissonance	SyntaxSemanticDissonanceType	Syntactically valid but semantically incorrect; syntactically valid but semantically incomplete; renderable but domain-inconsistent; textually valid PlantUML with wrong UML relation; syntactically invalid but semantically partially recoverable; other; not applicable.
Syntax-semantic dissonance	DissonanceEvidenceBasis	Short textual explanation of the evidence supporting the classification.
Limitations	Construct and evaluation threats	Author reported and reviewer inferred limitations, kept in separate fields

1.12.3 Missing and ambiguous data

The extraction database distinguishes *not reported*, *not applicable*, *unclear*, and *not accessible*. Reviewers must not infer missing model versions, prompts, parameters, metrics, or evaluator qualifications from external assumptions. Supplementary material and replication packages may be used when explicitly linked to the report. Authors may be contacted once for clarification when the missing information is essential to quality-evidence interpretation, and every contact attempt is logged.

1.12.4 Extraction pilot and verification

The forms will be piloted on a temporally and methodologically diverse sample of at least ten studies. The sample must include different UML diagram types and evaluation references. The pilot assesses field clarity, extraction time, category overlap, missing data frequency, and the feasibility of unit level extraction. All interpretive quality-evidence fields are independently extracted by two reviewers. Objective mapping metadata may be extracted by one reviewer and verified through an independent audit of at least 20 percent. Any material revision to the form or codebook triggers retrospective review of studies already extracted.

1.13 Methodological and Reporting Appraisal

Included studies are not excluded solely because of methodological weakness. Exclusion based on a methodological quality score could distort the map of an emerging area. All included studies are appraised to qualify the credibility and comparability of their quality evidence. Appraisal distinguishes methodological adequacy, reporting completeness, and availability of replication material. No single total score will be used as a substitute for domain specific judgment.

Table 1.21: Methodological and reporting appraisal domains

Code	Appraisal question	Judgment
A1	Are the study objective and research questions explicit and consistent with the evaluation?	Yes, Partially, No, Not applicable
A2	Are the requirements, domain material, dataset provenance, sample size, and selection procedure described?	Yes, Partially, No, Not applicable
A3	Are the LLM identity, version, access mode, instructions, context, tools, parameters, and execution date sufficiently reported?	Yes, Partially, No, Not applicable

Code	Appraisal question	Judgment
A4	Is the UML diagram type and output representation clearly specified?	Yes, Partially, No, Not applicable
A5	Are the quality constructs defined and operationalized?	Yes, Partially, No, Not applicable
A6	Is the evaluation reference explicit, inspectable, and justified?	Yes, Partially, No, Not applicable
A7	Are metrics, rubrics, units, thresholds, and aggregation rules sufficiently defined?	Yes, Partially, No, Not applicable
A8	Are evaluator qualifications, training, independence, and disagreement procedures reported when human judgment is used?	Yes, Partially, No, Not applicable
A9	Is stochastic variation addressed through repeated generation, sampling control, or uncertainty reporting?	Yes, Partially, No, Not applicable
A10	Are baselines and comparison conditions appropriate to the claim?	Yes, Partially, No, Not applicable
A11	Are quantitative or qualitative analyses appropriate and traceable to the data?	Yes, Partially, No, Not applicable
A12	Are threats, limitations, missing data, and failed cases reported?	Yes, Partially, No, Not applicable
A13	Are prompts, outputs, datasets, code, rubrics, or replication materials available?	Yes, Partially, No, Not applicable

Four domains are treated as critical for interpreting quality-evidence findings: A3, A5, A6, and A7. Sensitivity analyses compare conclusions derived from all included studies with conclusions derived from studies that have no *No* judgment in the critical domains. The appraisal is not used to manufacture a universal quality rank among heterogeneous study designs.

1.14 Coding and Taxonomy Construction

The taxonomy is constructed through a hybrid deductive and inductive process. The upper level structure is deductively informed by conceptual modeling quality and the UML specification Lindland et al. [1994], Krogstie et al. [2006], Object Management Group [2017]. Diagram specific categories and terminology are developed inductively from primary studies through thematic synthesis and constant comparison Cruzes and Dybå [2011], Krippendorff [2018].

1.14.1 Coding dimensions

Every extractable inadequacy is represented, where evidence permits, as a tuple

$$I = \langle R_v, O_d, C_u, S_e, E_x \rangle, \quad (1.1)$$

where R_v is the violated reference, O_d is the discrepancy operation, C_u is the UML carrier, S_e is the reported or inferred severity, and E_x is the example or evidence excerpt.

The initial dimensions are the following.

- i) **violated reference**, comprising concrete representation, UML abstract syntax, UML construct semantics, domain or requirements semantics, cross diagram consistency, or intended engineering task;
- ii) **discrepancy operation**, comprising omission, unsupported addition, incorrect representation, incorrect classification, inconsistency, duplication, redundancy, or emergent category;

- iii) **UML carrier**, comprising the affected diagram, element, relation, property, constraint, sequence, state, transition, message, behavior, or other type specific unit;
- iv) **severity or consequence**, preserved when the study reports a severity scale or task effect;
- v) **provenance**, comprising the original term, definition, page, table, figure, metric, or example supporting the code.

1.14.2 Coding procedure

- 1) Extract original terminology and definitions without immediately replacing them with normalized labels.
- 2) Apply the initial deductive dimensions to a pilot subset.
- 3) Create open codes for categories that cannot be represented without loss.
- 4) Compare codes across studies and diagram types, merge synonymous codes, separate overloaded codes, and document boundaries with positive and negative examples.
- 5) Construct diagram specific subcategories under shared transversal dimensions.
- 6) Search for negative cases that challenge category definitions.
- 7) Independently recode the pilot subset with the revised codebook.
- 8) Apply the stable codebook to the remaining included corpus.
- 9) Retrospectively recode earlier studies whenever a material category or boundary changes.

The codebook is versioned. Each code contains a definition, inclusion rule, exclusion rule, positive example, negative example, parent category, allowed UML carriers, and revision history. Original study labels remain available in a separate field so that normalization is fully auditable.

1.14.3 Coding reliability

Coding an inadequacy is two operations, and only the second carries an agreement coefficient. The first is unitization. The inadequacies reported by a study are enumerated in the study's own vocabulary before any normalized label is applied, and two reviewers may enumerate different numbers of them from the same study. The enumerated lists are therefore reconciled by discussion and adjudication before classification begins, and the reconciliation is recorded as the count of items added, removed, and merged on each side. Enumeration disagreement is reported as that count and is never absorbed into a coefficient computed over classification.

The second operation is classification. Two reviewers independently classify every item of the reconciled list, for every included study. Agreement is assessed at the normalized category level and at the principal dimensions of the tuple in Equation 1.1. Cohen kappa is used for binary and mutually exclusive nominal decisions Cohen [1960]. Krippendorff alpha is preferred when categories are multiple, data are missing, or more than two coders are involved Krippendorff [2018]. Reliability values are interpreted together with confusion matrices, category prevalence, and documented disagreement patterns. Disagreements are resolved by discussion and adjudication, and the consensus code is stored separately from the original reviewer codes.

Reconciling the units before classifying them is what makes a contingency table exist. An inadequacy is a repeatable field of free enumeration, so without a shared list of items there is no pairing between the two reviewers' codes, and a coefficient computed over unpaired enumerations would report how much each reviewer found rather than how far they agree on category

boundaries. The rule is fixed here, before the two independent passes, because aligning the units afterwards would let each reviewer’s classification decide which items are held to have been reported at all.

1.15 Synthesis Plan

1.15.1 Descriptive mapping of included studies

The included studies are mapped through frequencies, proportions, temporal distributions, cross tabulations, and co occurrence matrices. Planned analyses include the following.

- publication year by venue and publication type;
- diagram type by generation task;
- diagram type by input source and output representation;
- model family and version by year;
- context strategy by diagram type and task;
- study design by contribution type;
- quality dimension assessed, as given by the three evidence axes and by the dimension partition of the normalized construct, by diagram type;
- baseline and human involvement by study design;
- replication material availability by year and venue;
- publication status and methodological characteristics.

Counts are reported using both reports and unique primary studies where relevant. Categories with multiple values are explicitly identified as non exclusive. No meta analysis is planned because heterogeneous tasks, units, references, and metrics are expected. Any later quantitative aggregation requires a protocol amendment and a demonstrated basis for commensurability.

1.15.2 Extraction-based synthesis of quality constructs

Definitions are synthesized through a construct matrix containing the original term, reported definition, theoretical source, normalized construct, evaluation reference, unit, indicator, metric, and study context. The synthesis will identify convergent definitions, overloaded terms, unnamed operationalizations, and cases in which identical labels denote different procedures.

1.15.3 Taxonomy of reported UML inadequacies

The taxonomy is presented at two levels. The transversal level organizes discrepancy operations and violated references that recur across diagram types. The specific level records how these operations manifest in actors, use cases, classes, relations, attributes, messages, lifelines, states, transitions, activities, and other UML elements. Frequencies are presented as frequencies of reporting and coding within the reviewed corpus, not as estimates of real world defect prevalence unless the primary data support that inference.

1.15.4 Metric and evaluation-reference catalogue

Metrics and procedures are organized by the chain

$$\text{construct} \rightarrow \text{reference} \rightarrow \text{unit} \rightarrow \text{indicator} \rightarrow \text{procedure} \rightarrow \text{aggregation.} \quad (1.2)$$

The synthesis compares parsability, compilation, UML rule checks, precision, recall, F scores, coverage, edit distance, graph similarity, expert ratings, LLM judge scores, task performance, time, and other measures only after reconstructing their definitions and denominators. Measures with the same name but different operationalizations are not pooled.

1.15.5 Syntax-semantic dissonance synthesis

Syntax-semantic dissonance is analyzed from the explicit extraction fields for syntactic evidence, semantic evidence, dissonance classification, dissonance type, and evidence basis. The synthesis distinguishes direct evidence from indirect evidence and does not infer semantic adequacy from parseability, renderability, textual validity, or UML syntactic conformity alone.

Analytical subsets include studies with syntactic evidence only; studies with semantic evidence only; studies with both syntactic and semantic evidence; studies with direct syntax-semantic dissonance evidence; studies with indirect or non-assessable dissonance evidence; and studies generating PlantUML code. These subsets are derived during extraction and are not eligibility layers.

1.15.6 Generation context knowledge synthesis

This synthesis covers the whole eligible corpus, because what knowledge is supplied as generation context is a property of every generation setting and not only of the studies that also measured use. The synthesis examines how requirements, technical specifications, domain knowledge, UML technical instructions, prompts, examples, retrieval, tools, parser or renderer feedback, and iterative workflows are provided as generation context, and reports the supply of domain knowledge and the supply of the UML technical contract as the two axes of the resulting map.

1.15.7 Pragmatic adequacy and rework synthesis

This synthesis covers only the studies for which axis U is not absent, because a pragmatic or rework effect exists only where use was assessed, and running it over the whole corpus would read the silence of the studies that never measured use as an absence of effect. Reported context factors are classified as merely described, experimentally manipulated, statistically associated, or proposed without empirical evaluation, and causal language is used only when the original design supports it. Pragmatic and rework evidence is synthesized by task, actor, diagram type, outcome, and measure. A study that reports comprehensibility without an engineering task is not automatically coded as evidence of pragmatic adequacy.

1.15.8 Heterogeneity and sensitivity analysis

Heterogeneity is examined across diagram type, educational or professional context, input type, dataset provenance, model access, model family, publication status, evaluation reference, and appraisal domains. Sensitivity analyses will compare, where the corpus permits, peer reviewed versus preprint studies, expert reference versus LLM judge evaluation, repeated versus single generation studies, and stronger versus weaker reporting in critical appraisal domains.

Table 1.22: Reviewer roles

Role	Responsible searcher	re-	Responsibilities
Primary reviewer	Helaine Barreiros		Search execution, record management, complete screening, extraction coordination, synthesis, and audit log
Second reviewer	Emanoel Barreiros		Independent calibration, sampled title and abstract screening, complete full text screening, quality-evidence extraction and coding
Adjudicator	Ivaldir Honório		Resolution of unresolved eligibility, extraction, appraisal, and coding disagreements
Methodological validator	[Name]		Review of protocol, search strategy, selection process, synthesis, and reporting alignment
UML domain validator	[Name]		Review of diagram terminology, codebook boundaries, UML carrier categories, and reference distinctions
LLM domain validator	[Name]		Review of LLM terminology, model configuration fields, and context categories

1.16 Reviewer Roles, Conflict Resolution, and Audit

Every decision remains attributable to an individual reviewer and date. Consensus decisions do not overwrite original independent decisions. The audit trail preserves both. Reviewers declare prior authorship or close collaboration with any candidate primary report. A reviewer does not make the final eligibility or appraisal decision for their own work.

1.17 Protocol Validation

The protocol will be validated before registration and production screening through five complementary procedures.

- 1) **Conceptual validation.** Experts in UML, LLMs, and empirical Software Engineering review the scope, definitions, questions, and extraction fields.
- 2) **Search validation.** The search strategy is tested against the seed set, inspected by a methodological reviewer, and revised when failures reveal avoidable retrieval gaps.
- 3) **Selection pilot.** Reviewers apply the criteria to a heterogeneous sample and revise ambiguous criteria and examples.
- 4) **Extraction pilot.** Both forms are applied to at least ten studies and revised for clarity, feasibility, and coverage.
- 5) **Synthesis pilot.** The initial coding architecture is applied to a subset covering multiple diagram types and evaluation methods. Categories are revised before full coding.

Validation results, reviewer comments, decisions, and protocol changes are stored in a validation report. The final validated protocol will receive a new stable version and a persistent identifier before production selection begins.

1.18 Threats to Validity and Mitigation

Table 1.23: Principal threats and mitigation strategies

Threat	Potential consequence	Mitigation
Search terminology	Relevant studies may omit LLM, UML, generation, or requirements terminology used in the query	Broad synonym blocks, model names, diagram type names, seed validation, manual venue search, and snowballing
Database coverage	Recent workshop or preprint studies may be missing from major indexes	Multiple indexes, direct digital libraries, arXiv, manual venue search, and final search update
Venue selection bias	A predefined manual list may overrepresent expected Software Engineering communities and omit interdisciplinary outlets	Automated sources define the principal candidate corpus; manual venues are complementary; tiered activation criteria, venue decision logs, and incremental yield reporting are used
Source redundancy or premature removal	Overlap may increase workload, whereas unsupported source removal may reduce recall	Provenance preserving deduplication, seed recovery analysis, source contribution matrices, prespecified retention rules, and amendment controlled source changes
Scope ambiguity	Generic diagrams or non requirements inputs may be inconsistently included	Operational definitions, examples, calibration, dual full text screening, and adjudication
Publication family duplication	Conference, preprint, and journal versions may be counted as independent evidence	Family identification, primary report selection, complementary extraction, and unique study counts
Construct heterogeneity	Identical labels may denote different quality constructs or metrics	Original term preservation, construct matrix, reference and unit extraction, and no pooling without commensurability
Taxonomic normalization	Reviewer categories may overwrite source meanings	Separate original and normalized fields, codebook provenance, negative cases, double coding, and retrospective recoding
Reviewer subjectivity	Eligibility, extraction, appraisal, or coding may vary by reviewer	Training, calibration, independent decisions, agreement analysis, adjudication, and retained audit trail
Reporting bias	Studies may omit unsuccessful generations, defects, parameters, or evaluator disagreement	Appraisal of reporting, missing data coding, artifact inspection, author contact when essential, and cautious inference
Rapid model evolution	Findings may become technologically dated	Exact model and version extraction, execution date coding, temporal analysis, and search update
Preprint inclusion	Non peer reviewed evidence may distort conclusions	Explicit status coding, separate analysis, and sensitivity analysis
LLM assisted review work	Automated assistance may introduce untraceable or fabricated decisions	Human responsibility for every inclusion, extraction, and code; logging of tools, prompts, outputs, and verification

Threat	Potential consequence	Mitigation
Selective protocol change	Criteria or outcomes may be changed after observing results	Prospective registration, versioned amendments, reasons and dates, and impact assessment

1.19 Data Management, Transparency, and Use of Artificial Intelligence

The replication package will contain the registered protocol, search strings, search dates, raw exports when licensing permits, manual-search source manifests, permissible source snapshots or reproducibility hashes, raw documentary inventories, normalized metadata, inventory-conflict records, screening decisions, full text exclusion reasons, publication family links, extraction forms, codebook versions, appraisal judgments, analysis scripts, figures, tables, and amendment log. Licensed database exports, publisher pages, and copyrighted full texts are redistributed only when their terms permit it; otherwise the package preserves the query, source URL, retrieval date, permitted audit metadata, and checksums or reproduction instructions sufficient to document the retrieval without republishing restricted content Rethlefsen et al. [2021].

LLM or automation tools may be used for clerical support such as formatting, candidate duplicate detection, translation of search syntax, or consistency checking. They will not make autonomous eligibility, extraction, appraisal, or taxonomy decisions. Every substantive decision remains attributable to a human reviewer. Tool name, version, purpose, prompt or rule, output, and verification procedure are logged. Model generated bibliographic data are verified against authoritative records before use.

The review uses published literature and does not initially involve human participants. If authors are contacted for clarification, correspondence is limited to study information and stored according to institutional data governance requirements. Reviewer conflicts of interest are documented.

1.20 Protocol Version History and Amendments

Pre registration revisions are recorded as version history. After protocol registration, every change is recorded as a formal amendment rather than silently incorporated as if it had been prespecified Shamseer et al. [2015]. Changes include research questions, scope, sources, search terms, criteria, reviewer procedures, extraction fields, appraisal, coding, or synthesis.

Table 1.24: Protocol version history and amendment log

Version and date	Changed element	Rationale	Expected impact
1.0, [Date]	Initial systematic mapping protocol draft	Establish the complete methodological architecture before validation	Baseline draft; not registered
1.1, 2026-08-10	Information source strategy, manual venue sets, and source retention rules	Separate automated corpus identification from manual coverage validation and make source decisions auditable	Improved coverage rationale and reproducibility; no change to review questions or analytical design

Version and date	Changed element	Rationale	Expected impact
1.2, 2026-08-10	Source contribution controls, validity threats, completion criteria, and validation checklist	Register the operational evidence required to retain, remove, or expand sources and venues	Makes venue selection bias, database redundancy, manual yield, and source decisions explicitly auditable
1.3, 2026-08-10	Scopus search validation and source-specific query specification	Register controlled S1–S4 sensitivity tests, preserve the five prespecified seeds, remove the ambiguous “foundation model*” term, and restrict the UML block to title, abstract, or author keywords in Scopus	Establishes Scopus search-string version 1.0 with a documented record-level rationale; no change to review questions, analytical design, or eligibility criteria
1.4, 2026-08-10	IEEE Xplore search validation and source-specific query specification	Register the invalid Structured Advanced Search implementation I0 and the controlled Command Search rounds I1–I3; establish the IEEE-specific seed denominator; reject the over-restrictive UML-field modification and accept the LLM-field restriction	Establishes IEEE Xplore search-string version 1.0 while preserving source-specific field semantics and the review’s conceptual search architecture
1.5, 2026-08-11	Manual venue inspection procedure	Specify the handsearching unit, source hierarchy, census-like title and abstract inspection, candidate statuses, required record fields, positive coverage controls, cross-source merge, incremental yield, completion rules, initial operational order, and audit outputs	Makes manual coverage validation reproducible and separates discovery from formal eligibility screening; no change to review questions, eligibility criteria, or final review interval

Version and date	Changed element	Rationale	Expected impact
1.6, 2026-08-11	ACM Full-Text Collection search validation	Register the ACM-specific seed denominator, interface field-prefix tests, A1–A3 controlled validation rounds, rejection of the UML-field restriction because of diagnostic false negatives, acceptance of the LLM-field restriction, cancellation of A4, and the exact ACM production query implementation	Establishes ACM Full-Text Collection search-strategy version 1.0 while preserving source-specific field behavior and the review’s conceptual search architecture
1.7, 2026-08-11	Manual-search provenance and documentary-inventory controls	Formalize source roles, PRIMARY_TOC membership, source manifests, permissible snapshots and hashes, immutable raw documentary inventories, raw-to-normalized identifier traceability, InventoryConflict reconciliation, operational unit states, and strengthened COMPLETE criteria	Makes the manual-search population and every transformation from documentary source to screening decision independently auditable; no change to research questions, eligibility criteria, analytical design, or review interval
1.8, 2026-08-17	Screening gate redesign and eligibility criteria realignment (amendment A004)	Introduce the gate structure D, A1–A3, B0–B5, C1, with one question and at most one criterion per filter; renumber inclusion criteria I1–I8 in filter order while freezing exclusion codes; add E7b and E12; withdraw E5, E10, and E11 as declared gaps; convert full-text unavailability into the <i>identified, not retrieved</i> attrition stratum; replace the evidence filter with the extraction axes L, D, and U plus the result-attribution attribute; compute the analytical subsets from the axes	Removes the only eligibility filter that operated on the review’s dependent variable and the only criterion requiring exhaustive reading before it could be applied; no change to research questions, search strings, information sources, temporal and language scope, or the exclusion decisions already recorded

Version and date	Changed element	Rationale	Expected impact
[Validated version and date]	Initial registered protocol	Prospective registration after completion of the validation checklist	Baseline for post registration amendments

When a codebook change occurs during qualitative synthesis, the codebook version and retrospective recoding action are recorded in the analytical log. Such refinements do not automatically constitute changes to the review questions or eligibility criteria, but their effect on interpretation must remain auditable.

1.21 Reporting and Completion Criteria

The secondary study will be considered methodologically complete when the following conditions are satisfied.

- 1) The validated search has been executed in all retained automated sources and updated before final reporting.
- 2) The complete core manual venue set has been inspected, every conditional venue trigger has been resolved, and the incremental yield of manual searching has been reported.
- 3) Every required manual-search unit has a documented operational state; before final synthesis, every available unit has reached **COMPLETE**, while genuinely unavailable future proceedings remain explicitly **PENDING**. Source manifests, raw and normalized inventories, discovery logs, candidate sets, and coverage comparisons are preserved in the replication package.
- 4) Source contribution, redundancy, seed recovery, and any source retention or removal decision are documented.
- 5) Deduplication, selection, and publication family reconciliation are complete and auditable.
- 6) Every included study has a verified mapping and generation-context extraction record.
- 7) Every included study has independently verified quality-evidence extraction and appraisal.
- 8) The codebook has reached a stable version and all included studies have been coded under that version.
- 9) Research question specific syntheses have been completed and checked against the extracted data.
- 10) Sensitivity analyses and negative case checks have been performed where applicable.
- 11) The PRISMA flow, SEGRESS checklist, PRISMA-S search record, exclusion table, replication package, and amendment log are complete.
- 12) Conclusions distinguish the mapped corpus, extraction-derived analytical subsets, publication reports, unique primary studies, and evaluation instances.

The final report will avoid causal or prevalence claims that exceed the designs and units represented in the primary studies. Frequencies in the map describe the literature, not necessarily the real frequency of inadequacies in LLM generated UML diagrams. The taxonomy describes reported and operationalized inadequacies, not a complete ontology of every defect that can occur.

1.22 Protocol Validation Checklist

Table 1.25: Checklist for protocol validation before registration

ID	Validation item	Status
V01	The focused systematic mapping design and quality-evidence emphasis are explicitly justified.	[Pending]
V02	The general objective and every research question are approved by the review team.	[Pending]
V03	Operational definitions resolve the principal scope ambiguities.	[Pending]
V04	The unified eligibility criteria have positive and negative examples, one inclusion and one exclusion criterion per filter, and no filter operating on the review's dependent variable.	[Pending]
V05	Core and provisional information sources, source retention decisions, and the core and conditional manual venue sets have been validated and documented.	[Pending]
V06	The baseline string retrieves the complete seed set or every failure has a documented complementary route, and a source contribution matrix has been produced.	[Pending]
V07	Search strings for every source have been peer reviewed and preserved verbatim.	[Pending]
V08	Reviewer roles, independence, conflict resolution, and availability are confirmed.	[Pending]
V09	Selection calibration has met the predefined agreement target.	[Pending]
V10	Mapping, generation-context, and quality-evidence extraction fields have been piloted and revised.	[Pending]
V11	The appraisal instrument has been piloted across heterogeneous study designs.	[Pending]
V12	The taxonomy codebook has been piloted across multiple UML diagram types.	[Pending]
V13	Data management, backup, licensing, and replication package procedures are operational.	[Pending]
V14	The protocol has been checked against SEGRESS, PRISMA-P, PRISMA 2020, and PRISMA-S.	[Pending]
V15	Manual venue incremental yield and every conditional venue activation decision are documented.	[Pending]
V16	Manual-search units, discovery statuses, candidate rationales, cross-source matches, and completion states are represented in the manual-search audit log and tested on at least one core venue-year unit.	[Pending]
V17	The manual-search provenance chain has been operationally tested from PRIMARY_TOC through source manifest, raw inventory, normalized inventory, discovery status, and conflict reconciliation on at least one core venue-year unit.	[Pending]
V18	The validated version has been registered and assigned a persistent identifier.	[Pending]

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